

# Ecosystem carbon measurement in Grasslands



## Slide 1

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CW1

**Speaker: both/whoever**

Clare Wark, 2026-05-01T13:54:40.726

# Day 1 - Agenda

CW1

## Morning

- Introductions
- What is carbon?
- Why measure carbon?
- Tools and resources
- Examples of carbon projects
- Desktop analysis
- Sampling design

## Afternoon

- Equipment and datasheet overview
- Field methods for soils
- Testing out tools (outside)



## Slide 2

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CW1 @Cal, add in the new section on sampling design here when ready  
Clare Wark, 2024-10-04T14:00:24.305

CD1 0 Added under "implementing a carbon monitoring project"  
Cathal Doherty, 2024-10-10T18:04:42.623

# Day 2 - Agenda

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## Morning

- Implementing a carbon project (outside)

## Afternoon

- Implementing a carbon project (outside)
- Post-training survey
- Wrap up Q&A



Photo: Andrew Price



# Introductions

- Name
- Role
- Icebreaker question

CW1

CW1      **Do it irl since people will be introducing themselves anyways**

Clare Wark, 2024-10-01T17:09:38.595

CW1 0      **What season would you be and why? Unlimited item for rest of life? ...?**

Clare Wark, 2026-04-30T18:03:22.074



What is carbon?  
Why measure it?

CW1

## Slide 5

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CW1      **Make this a menti question for the group to contribute responses from their phones/laptops**  
Clare Wark, 2024-09-30T16:10:32.290

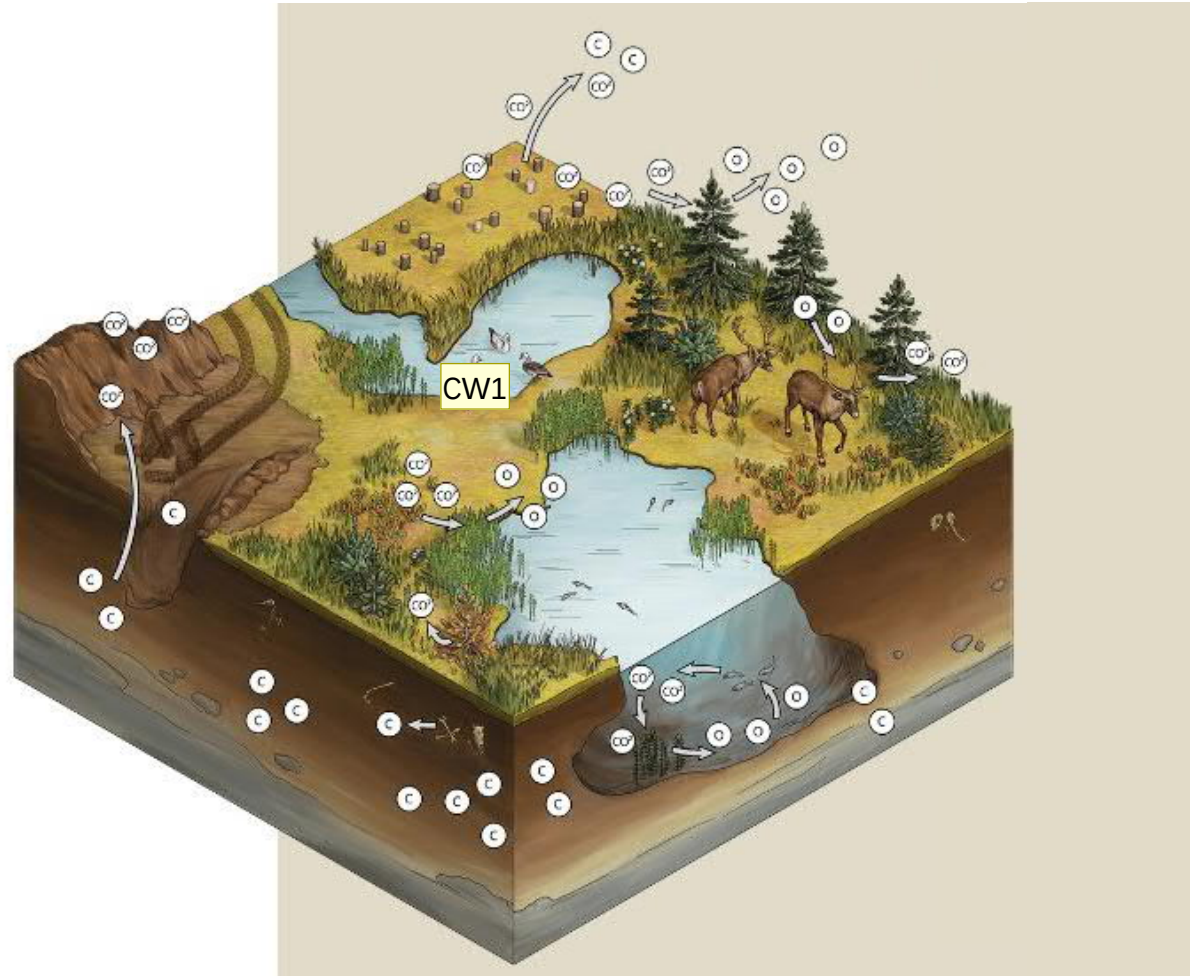
CW1 0    **Participation link:**  
**<https://www.menti.com/alrw2g1d5js2>**

**Slide 1**

Clare Wark, 2024-09-30T16:16:09.798

# What is carbon?

- A building block for all living things
- Carbon moves in a cycle



CW1      **Keep this in?**

Clare Wark, 2024-03-11T23:36:39.228

JC1 0      **You might want to change to trees, other vegetation and soils?**

Forests includes trees and soils, and aboveground biomass includes trees... so this breakdown is confusing.

I would just focus on trees and soils if that's your strength, no need to do other vegetation if you're not familiar with the sampling.

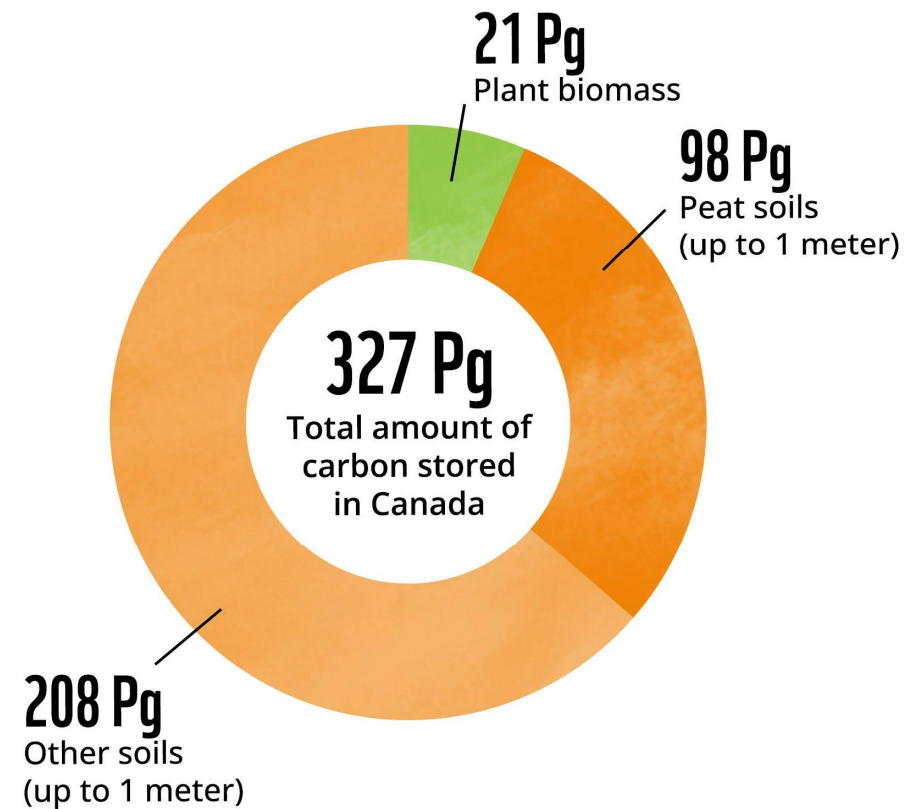
Jessica Currie, 2024-03-12T11:23:41.644

JC1 1      **You had "forests, above-ground biomass and soils"**

Jessica Currie, 2024-03-12T13:12:43.177

# Carbon in Canada

- An estimated 327 billion tonnes of carbon is stored in Canada's plants and soils
- This does not include the carbon found in marine and freshwater ecosystems



Source: Sothe et al. 2022

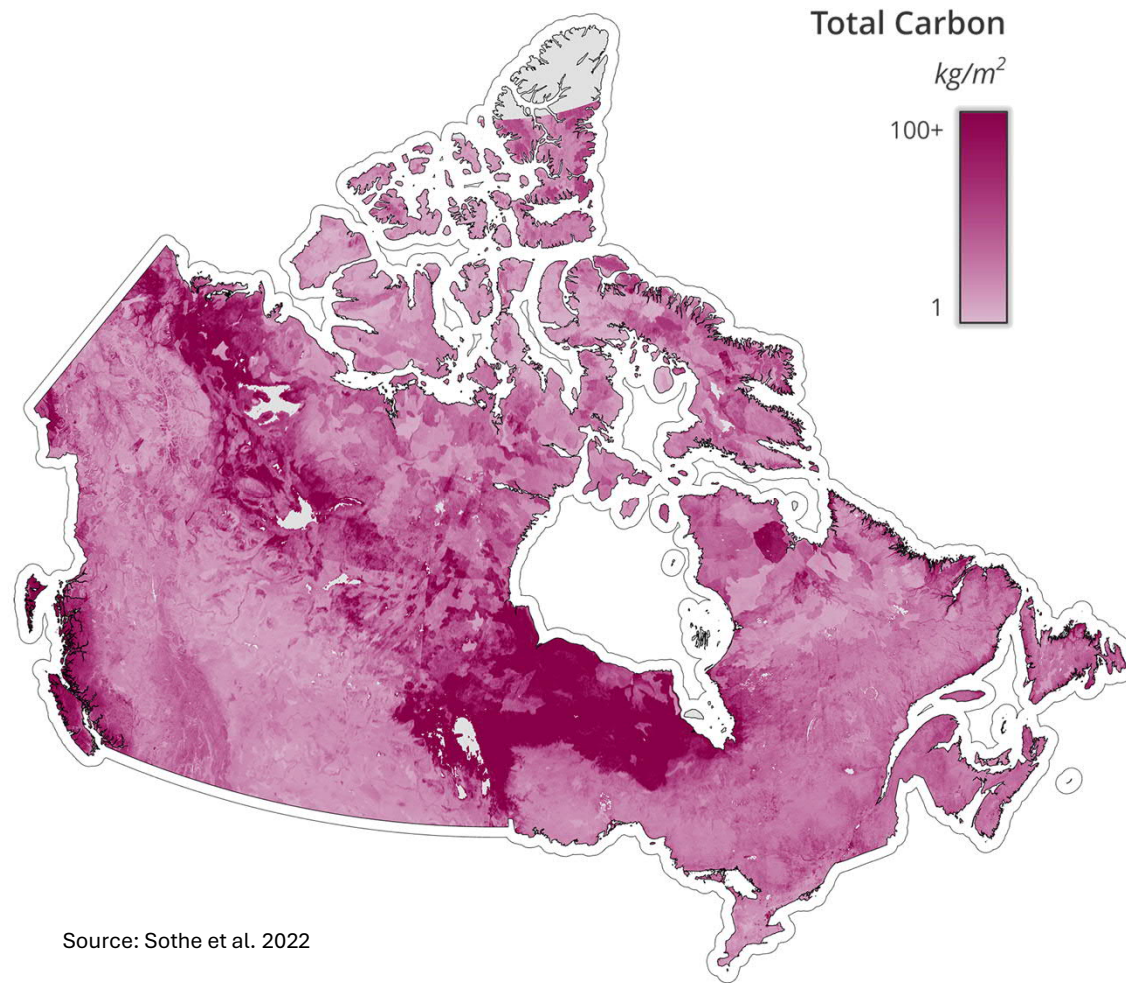
## Slide 7

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CW1      **Have them guess the pools corresponding to the circle chart**  
Clare Wark, 2026-05-01T15:00:26.941

CW1 0    **Reveal animation added**  
Clare Wark, 2026-05-01T15:00:37.482

# WWF-Canada carbon map



Source: Sothe et al. 2022

# Measuring carbon in forests and soils

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- Carbon measurement:
  - Collecting data in the field to estimate the amount of carbon stored in a specific carbon pool
- Carbon monitoring:
  - Ongoing measurements, assessments, or other forms of data collection

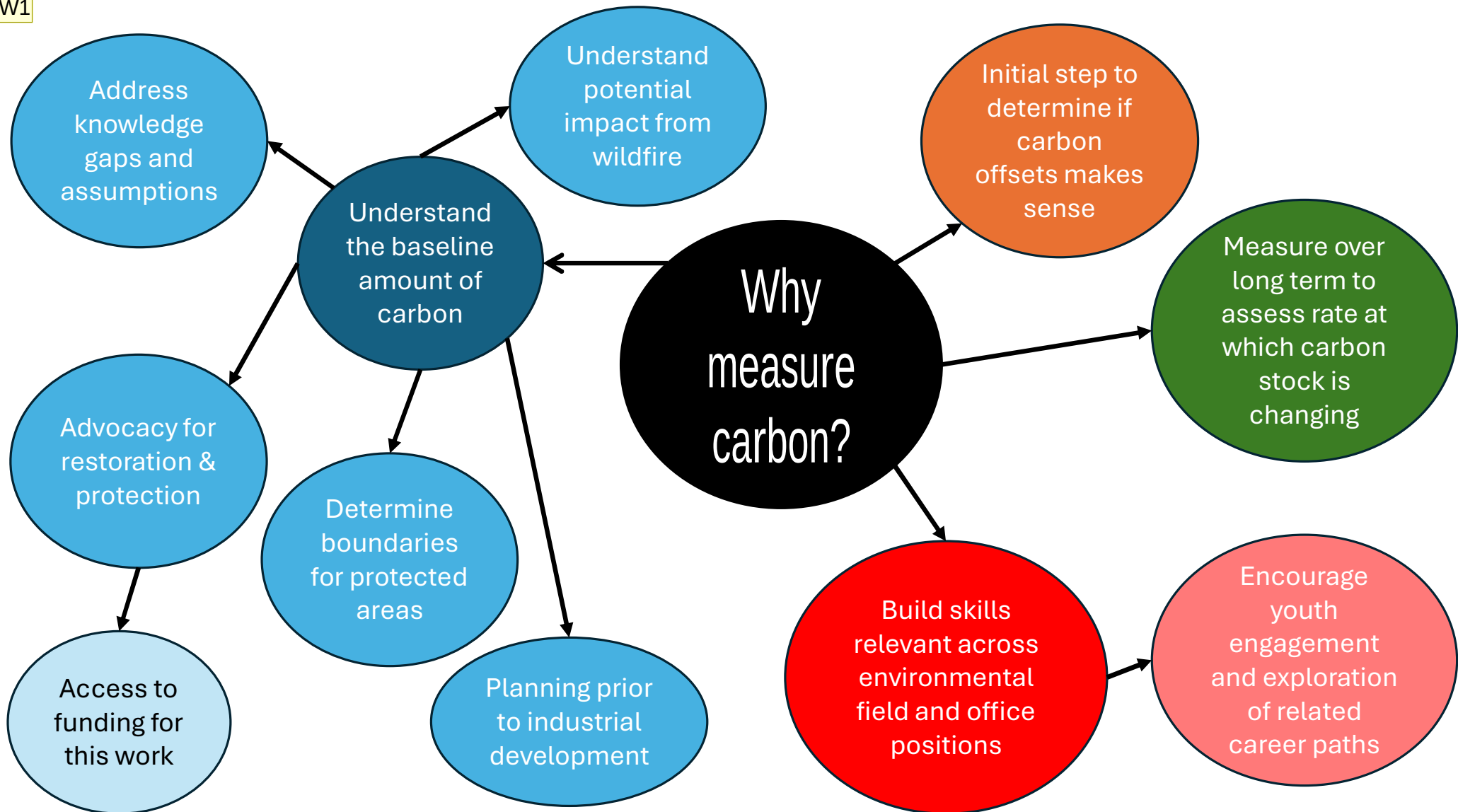


JC1

This is what you had before, but I believe the "eg's" are inaccurate. If you're soil sampling and want to assess carbon sequestration over time, you go back to the same sites year over year.

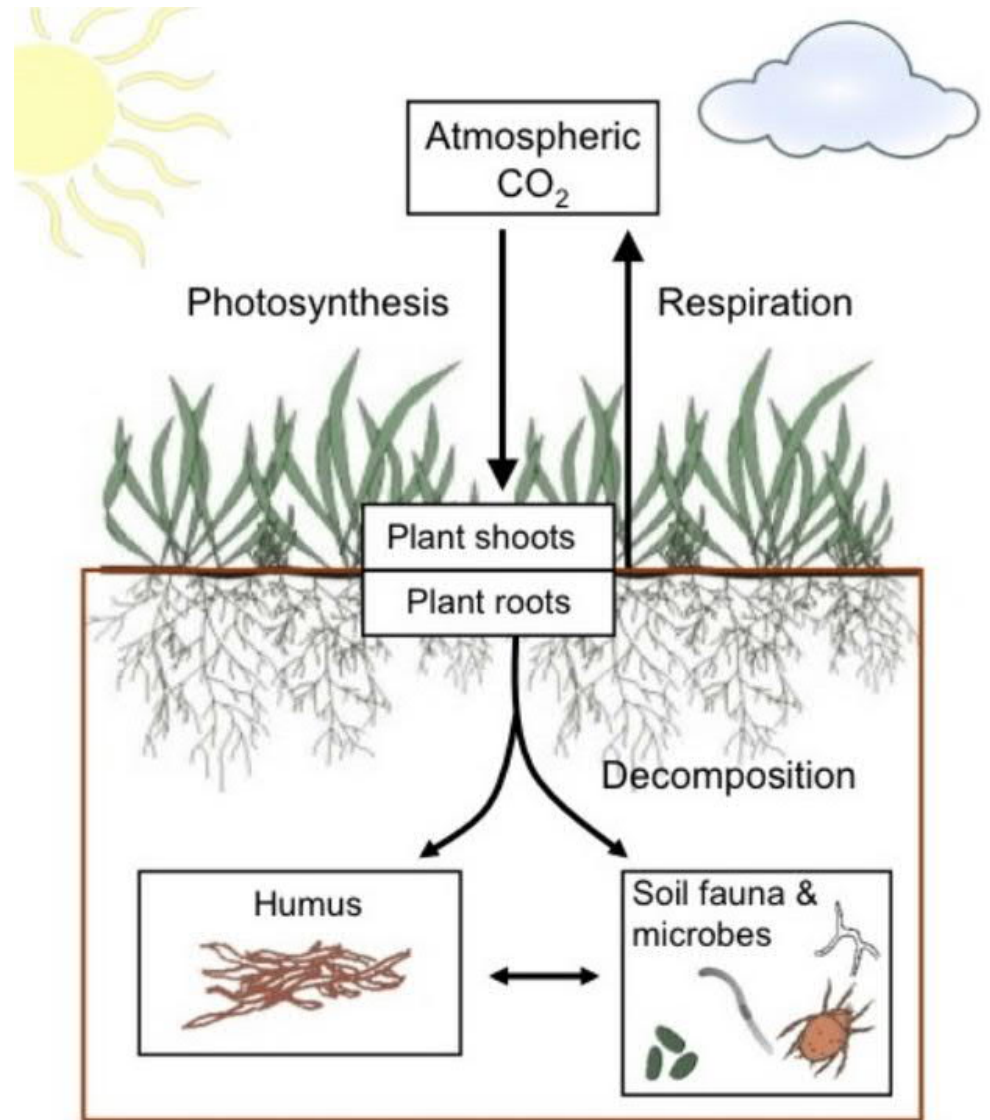
May involve returning to the same sites and plots year over year (e.g., a forest) or can include new sites and plots each year within a specific area (e.g.: soil cores)

Jessica Currie, 2024-03-12T11:26:52.034



CW1      **Ask group if they can think of other reasons to measure carbon**  
Clare Wark, 2026-05-01T19:11:29.699

Grassland  
ecosystems that  
capture and store  
carbon in plant tissue  
and in soils

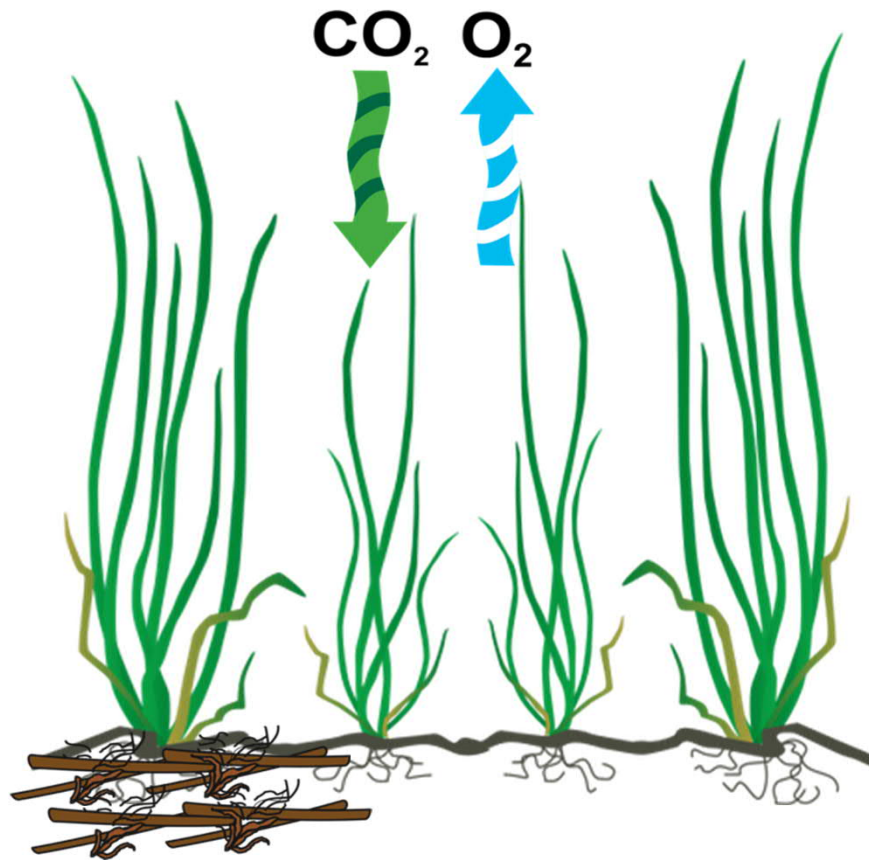


# What is carbon cycling?

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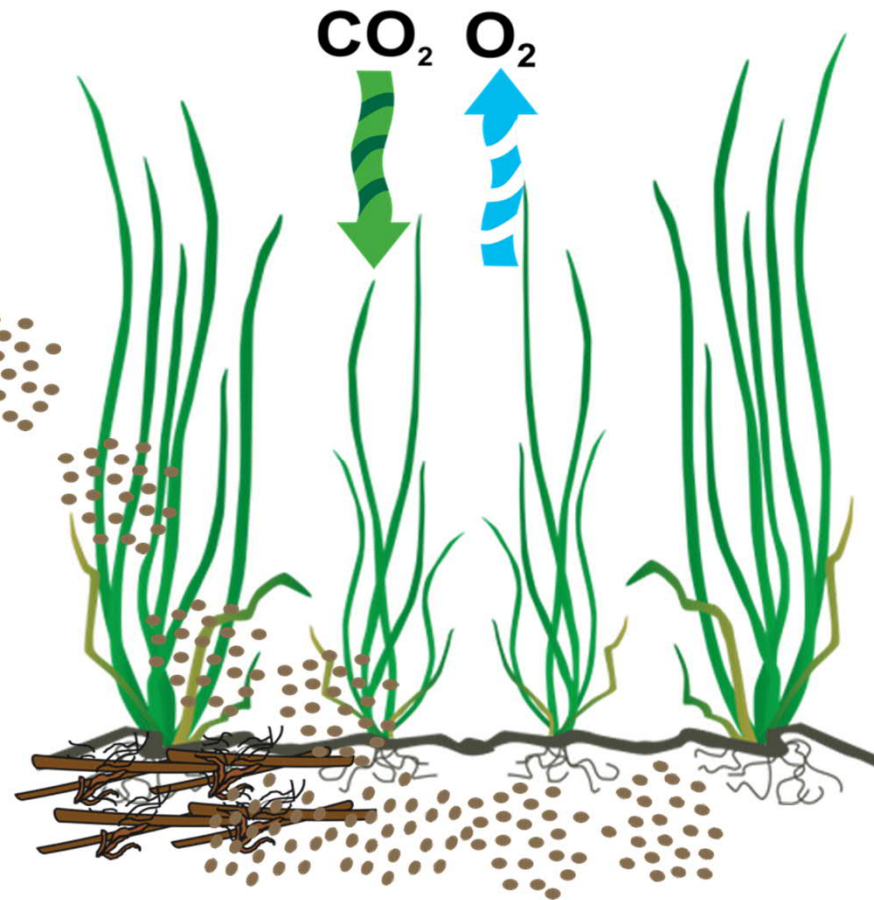
Plants are taking in  $\text{CO}_2$   
And breathing out  $\text{O}_2$

Carbon entering ecosystem  
via photosynthesis



# What is the carbon cycling?

Weathering and transport of suspended particles into the ecosystem



Brought in from tides or upstream from the watershed

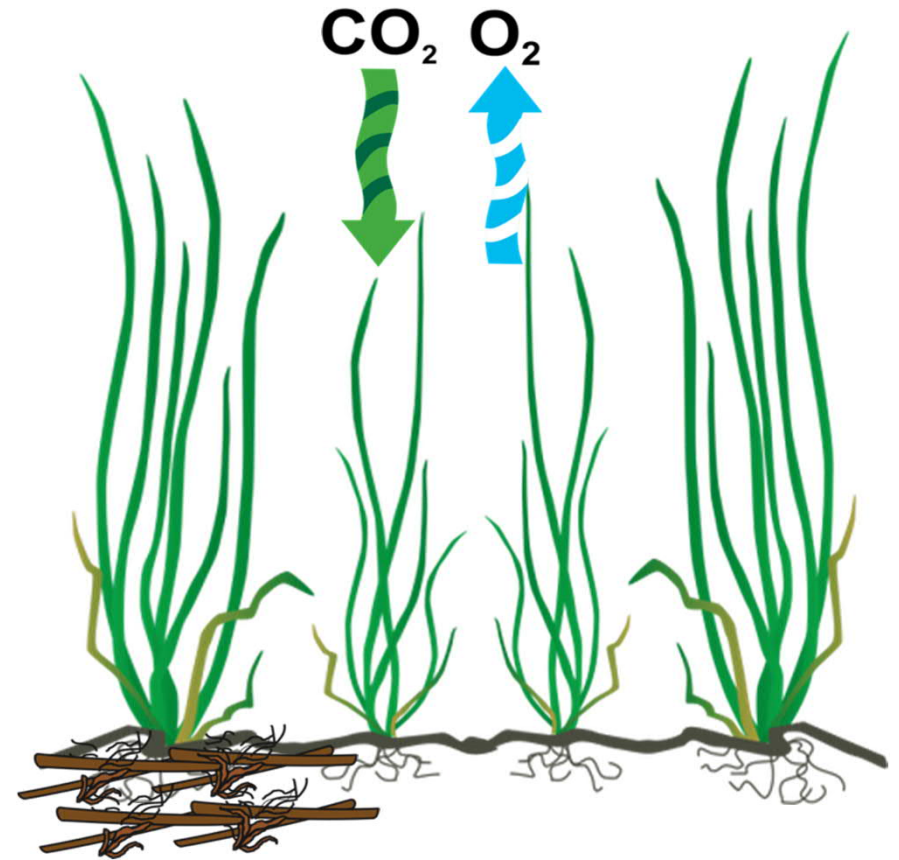
# Where does the carbon come from?

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Autochthonous (Produced in-place)  
Seagrass photosynthesis  
Root and rhizome inputs

Allochthonous (Transported in)  
From tides, rivers, adjacent systems  
Trapped in sediments

Seagrass meadows are both carbon  
producers and carbon traps.



# Where does the carbon come from?

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Seagrass photosynthesis

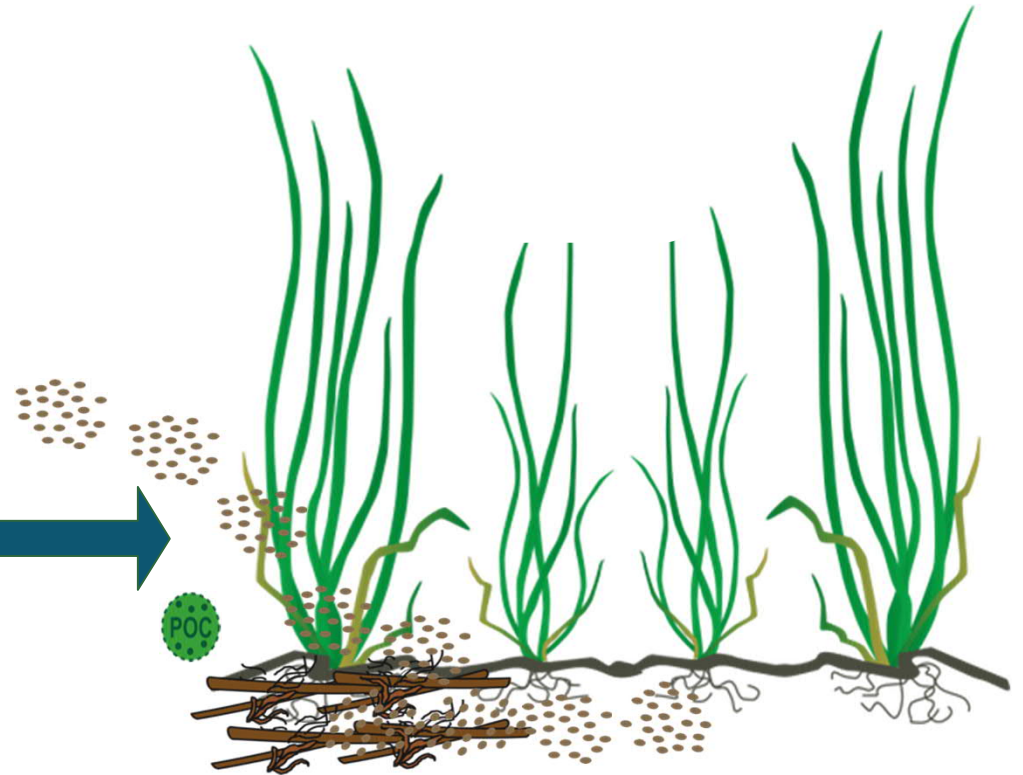
Root and rhizome inputs

Allochthonous (Transported in)

From tides, rivers, adjacent systems

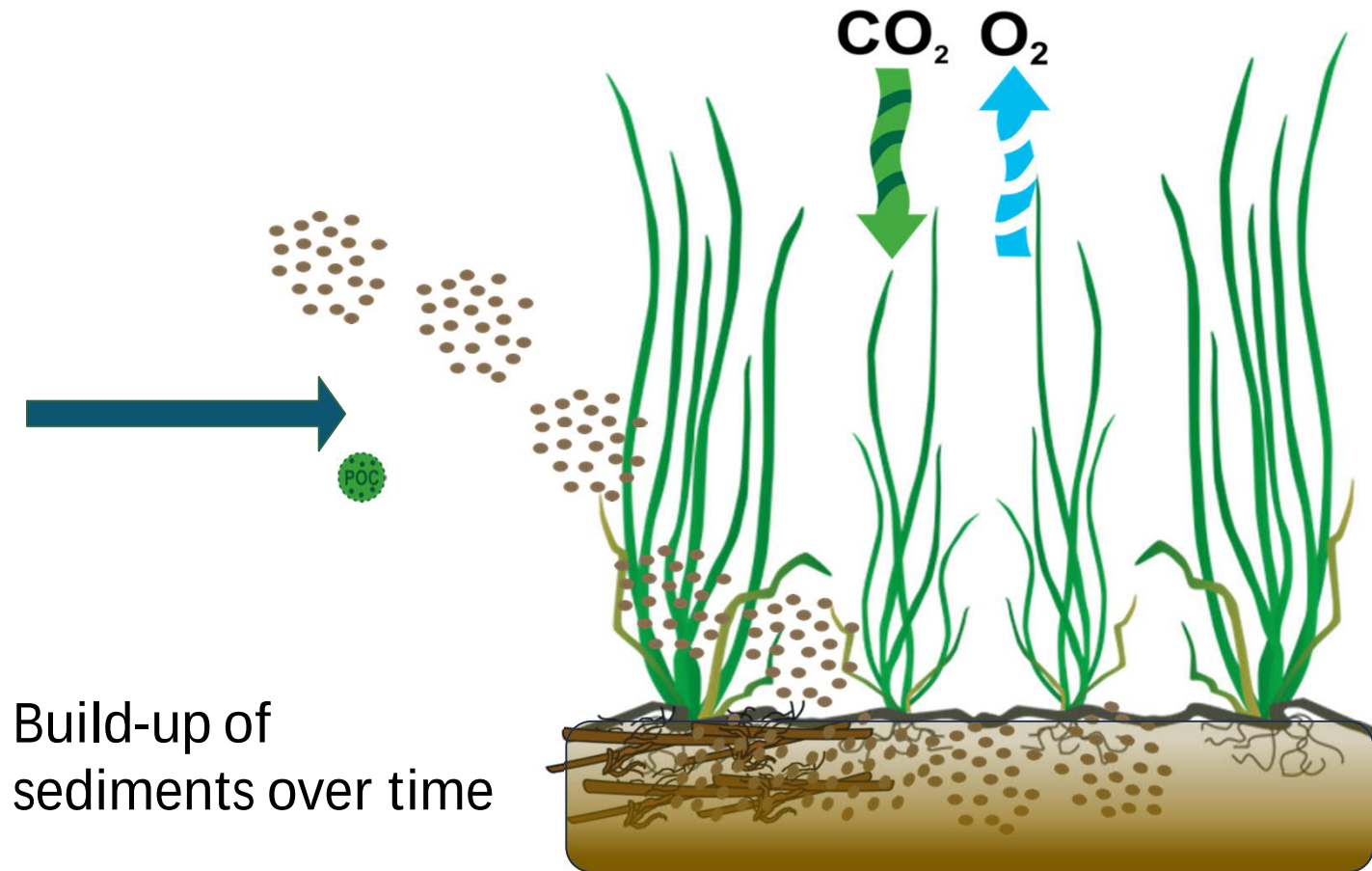
Trapped in sediments

Seagrass meadows are both carbon producers and carbon traps.



# What is the soil carbon cycling?

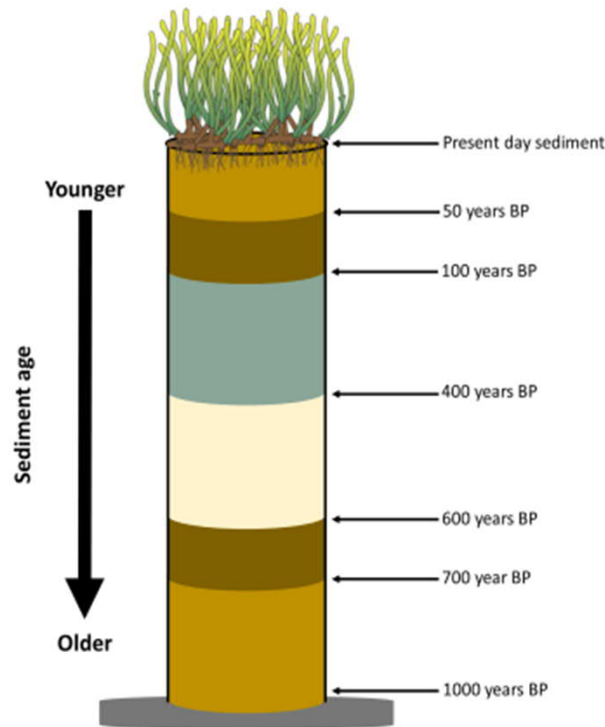
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# What is the soil carbon cycling?

## What is sediment?

- “Layers of **transported** and **deposited particles** or aggregates derived from rocks, soil, or biological material” (SSSA 2021)



# Carbon stocks vs carbon rates

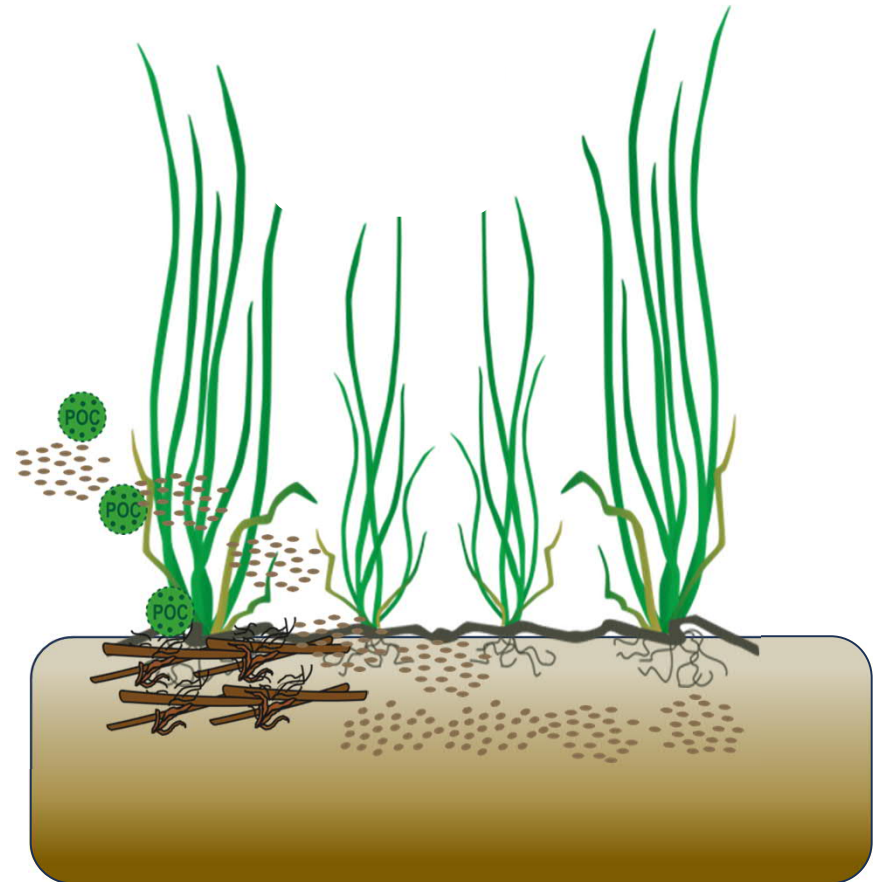
---

**Carbon Stock** Total carbon stored at one point in time

**Sequestration Rate** Carbon stored per year

Seagrass systems = high input and output.  
Net storage happens in sediments.

*Sediment cores measure stocks — but they reflect long-term system dynamics.*



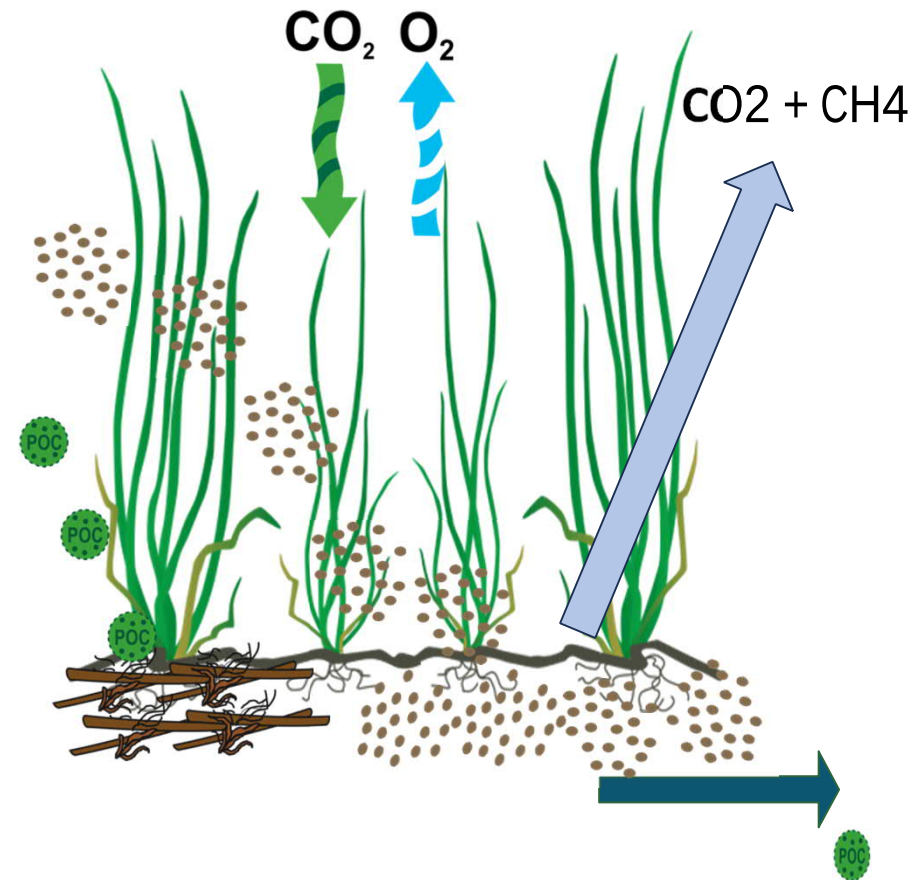
# Carbon stocks vs carbon rates

**Carbon Stock** Total carbon stored at one point in time

**Sequestration Rate** Carbon stored over time (year)

Seagrass systems = high input and output.  
Net storage happens in sediments.

*Sediment cores measure stocks — but they reflect long-term system dynamics.*



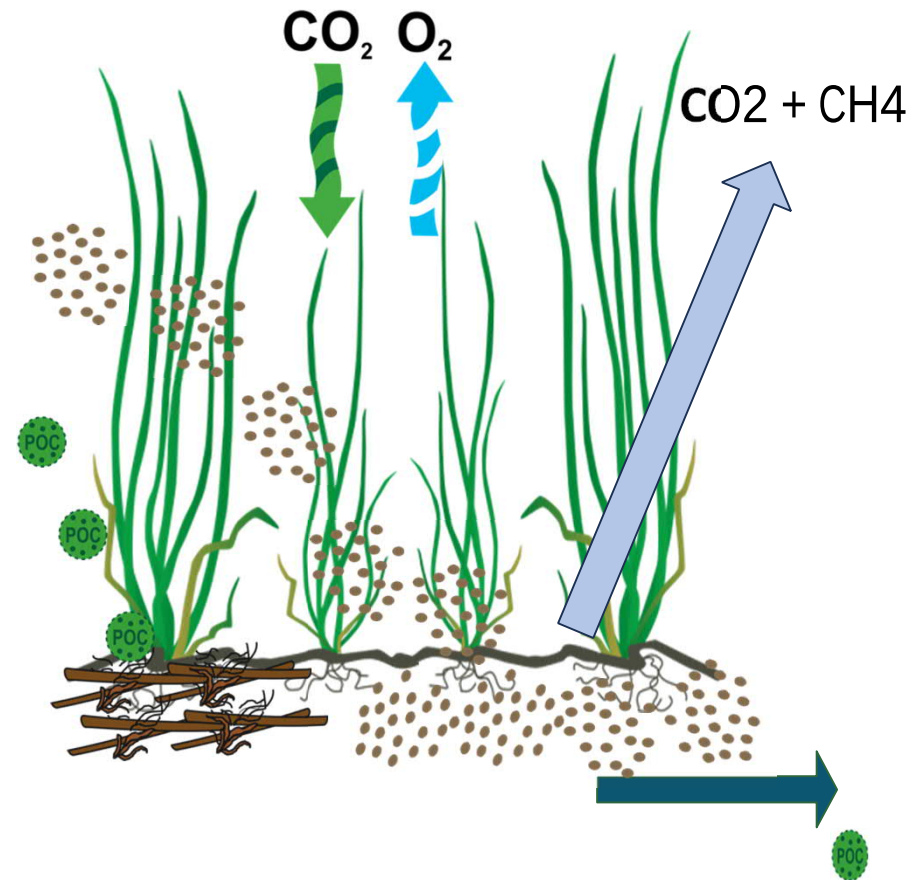
# Carbon stocks vs carbon rates

**Carbon Stock** Total carbon stored at one point in time

**“Fluxes” or “Processes” = Arrows**

Seagrass systems = high input and output. Net storage happens in sediments.

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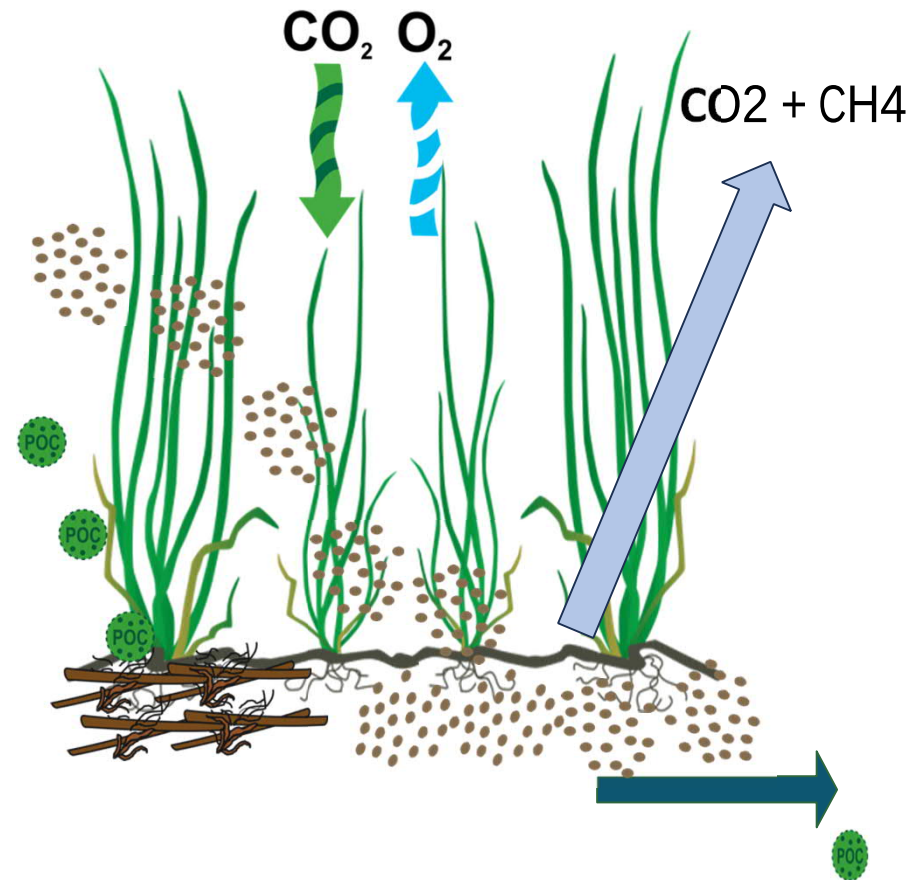
# Carbon stocks vs carbon rates

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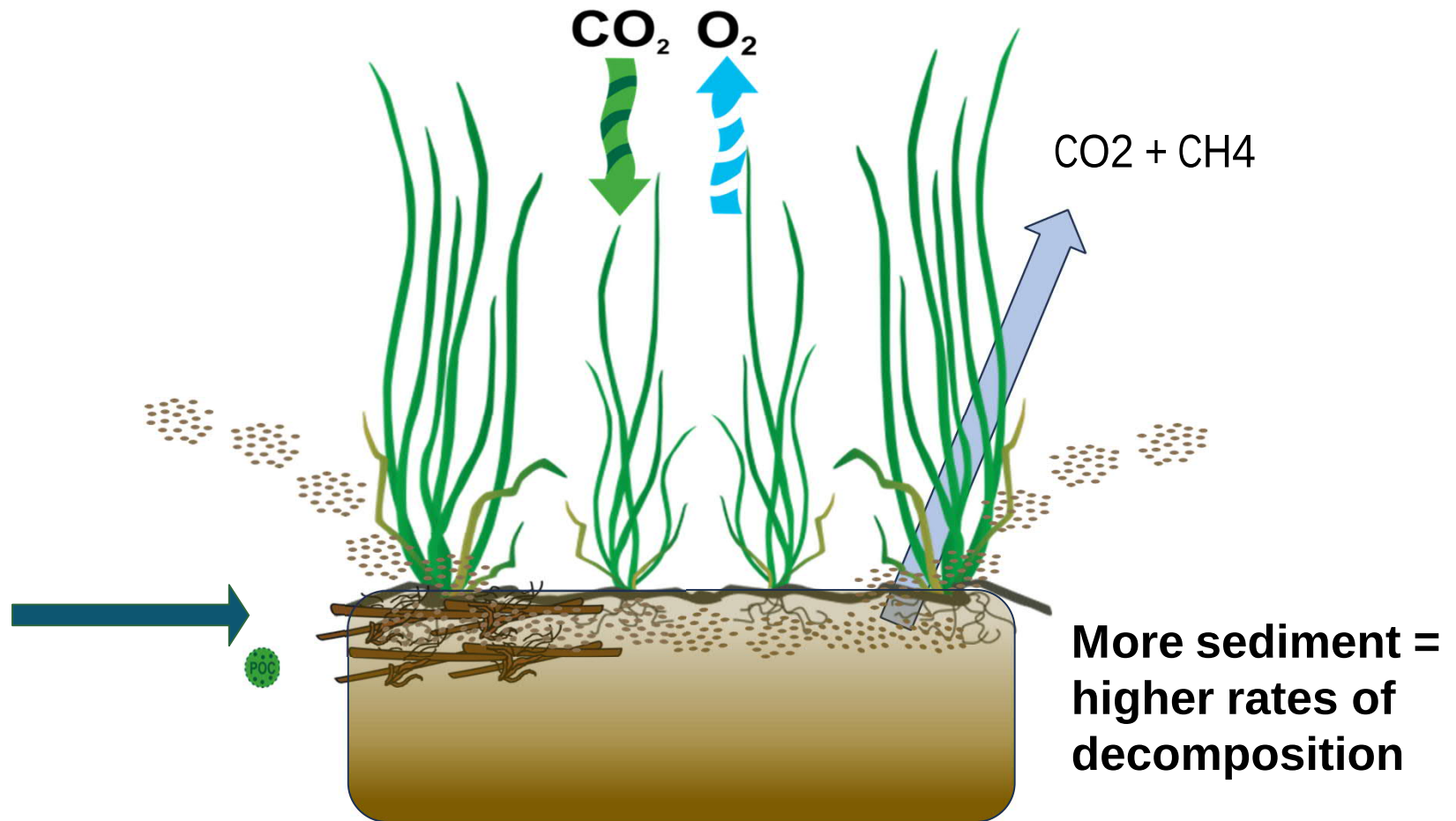
**“Fluxes” or “Processes” = Arrows**

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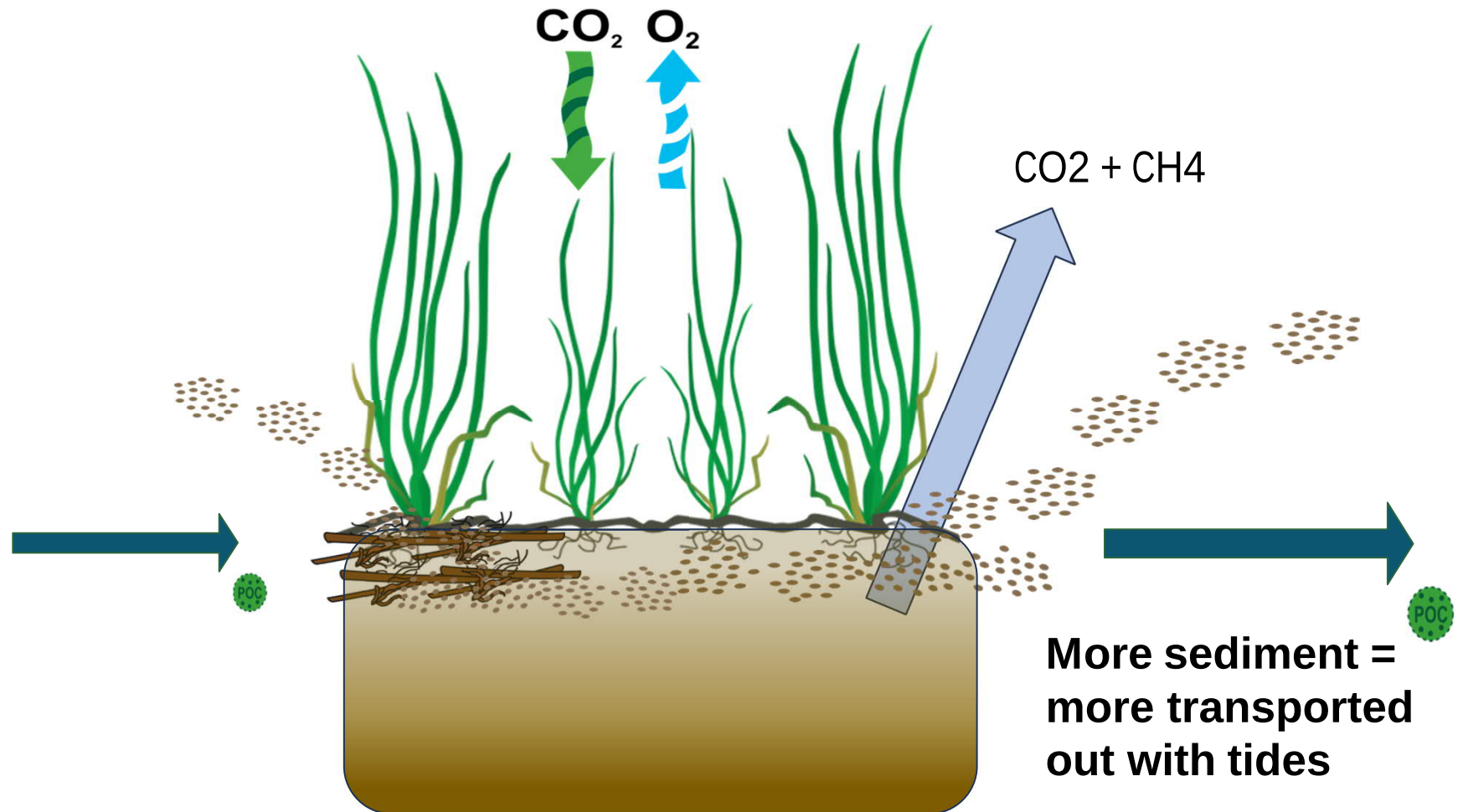
**Sediment cores** measure stocks which reflect long-term fluxes



# What is the soil carbon cycling?



# What is the soil carbon cycling?

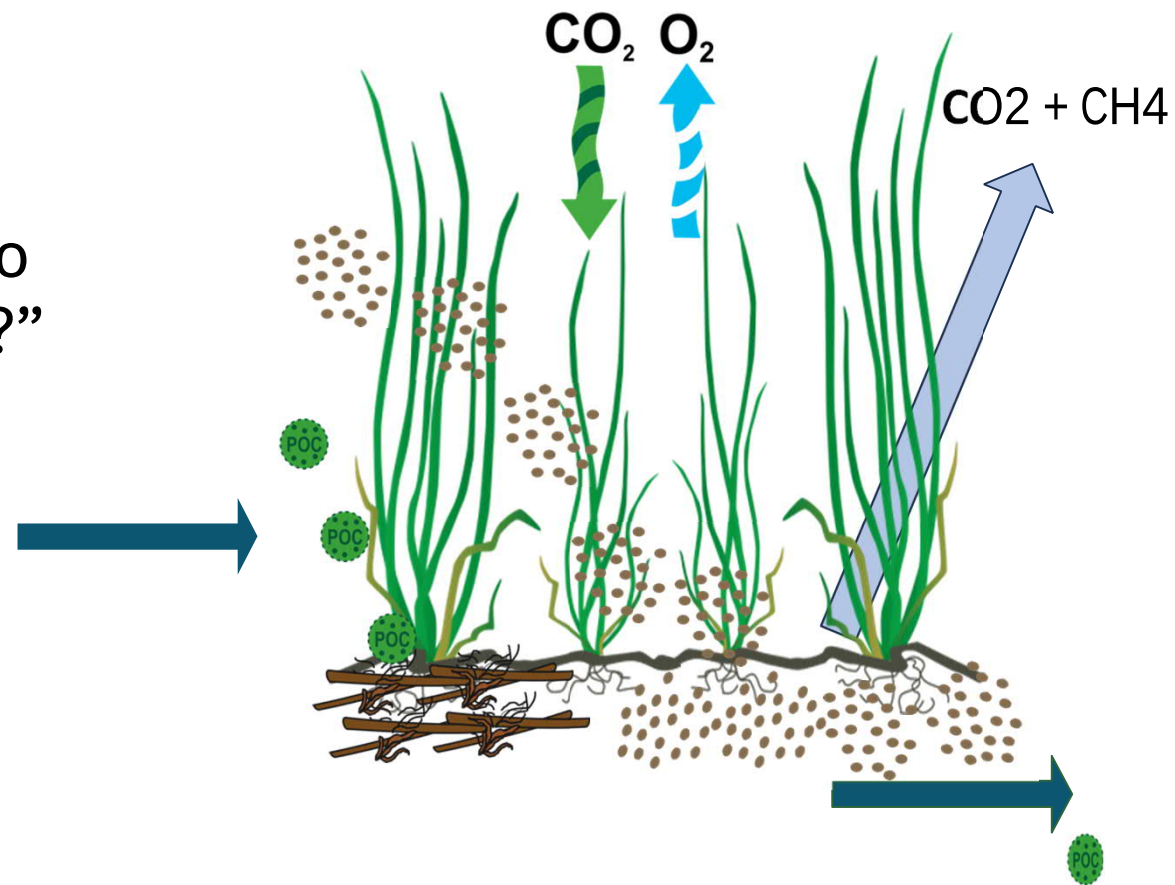


# What is the soil carbon cycling?



## Net balance

“Is more carbon being added to the soil than is being released?”



# Carbon balance = In - Out



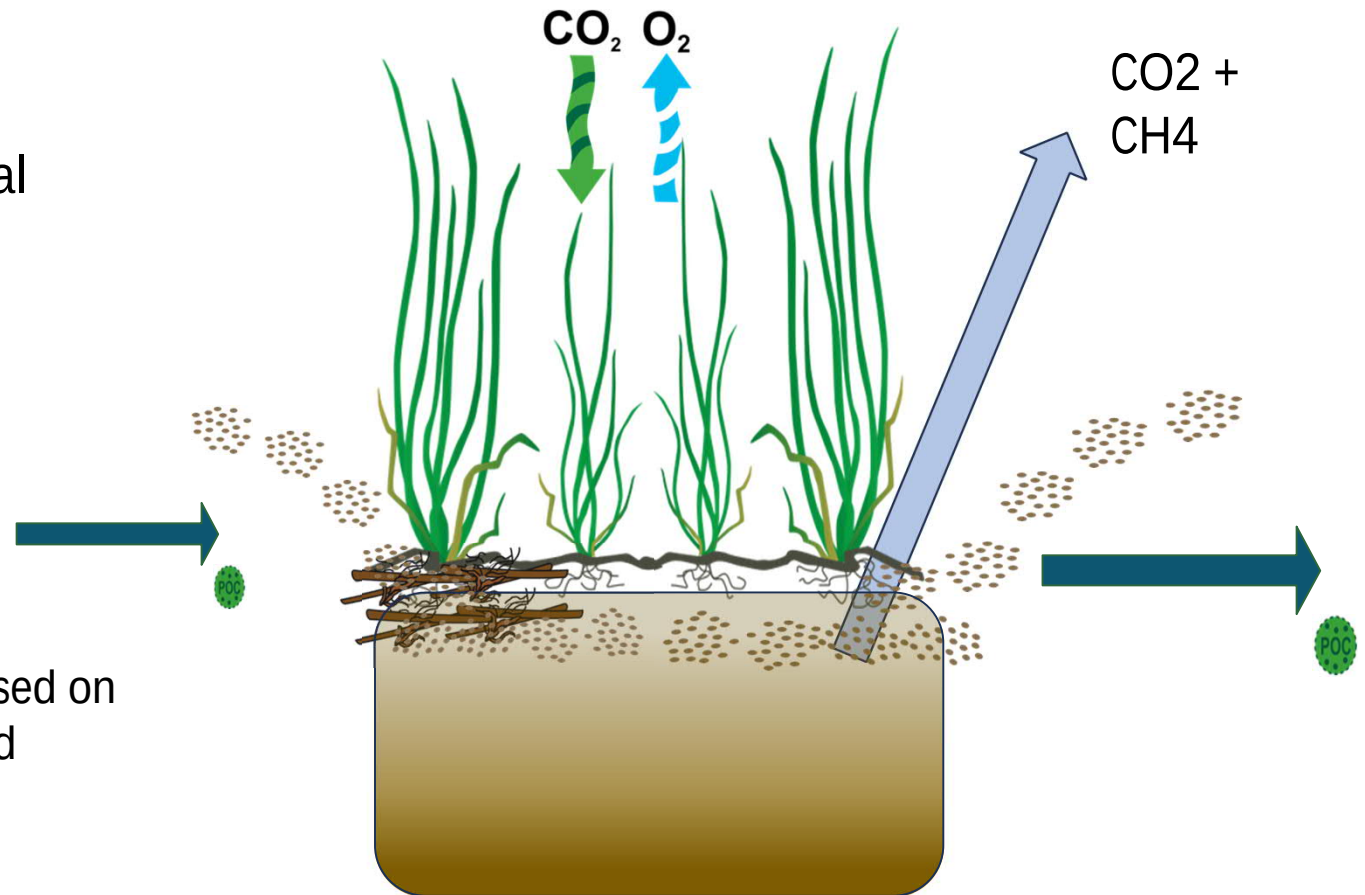
## Carbon IN

- Photosynthesis
- Transported organic material
- Sediment inputs

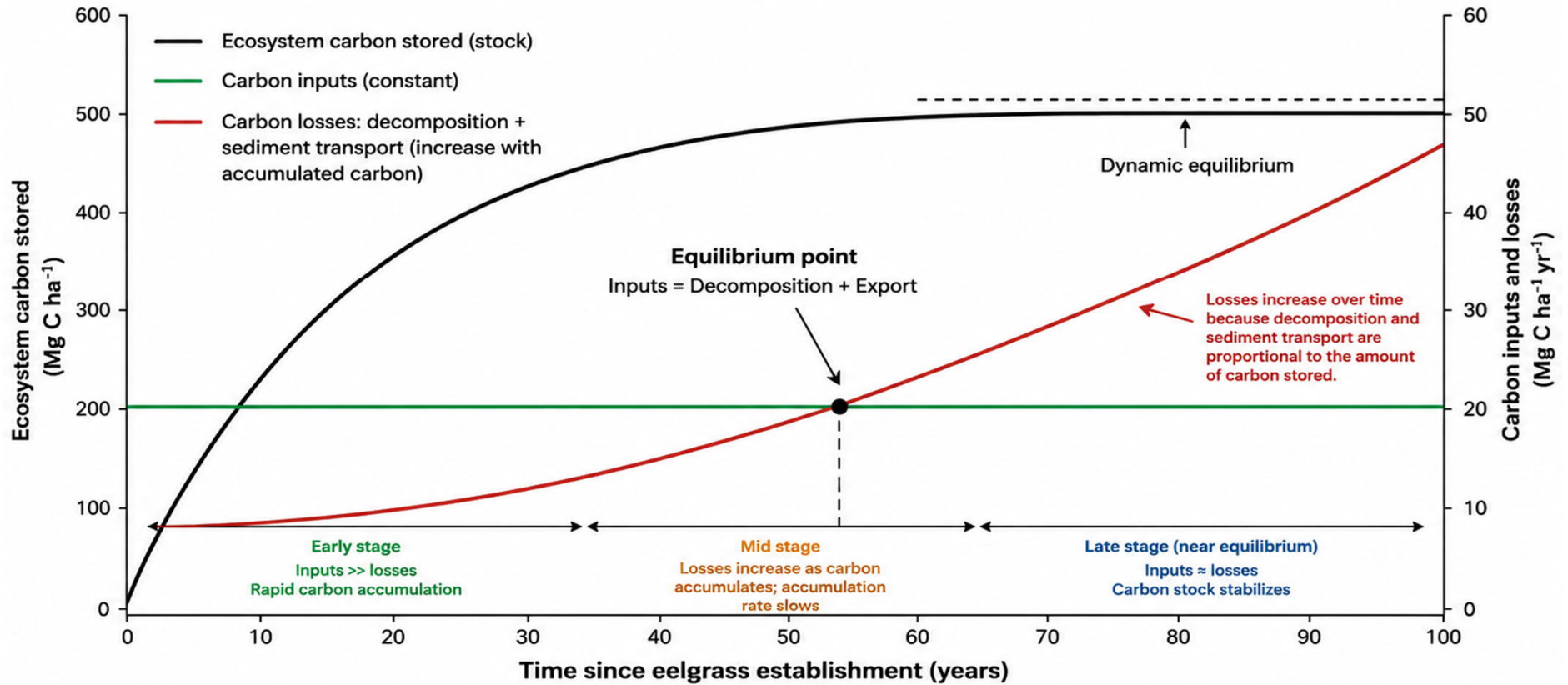
## Carbon OUT

- Respiration →  $\text{CO}_2$
- Methane ( $\text{CH}_4$ )
- Export via tides

Ecosystem carbon functions based on the balance between inputs and outputs.



## Carbon accumulation in an eelgrass ecosystem over time



### Carbon inputs (constant through time)

- Carbon fixation through photosynthesis
- Allochthonous carbon deposition from water column



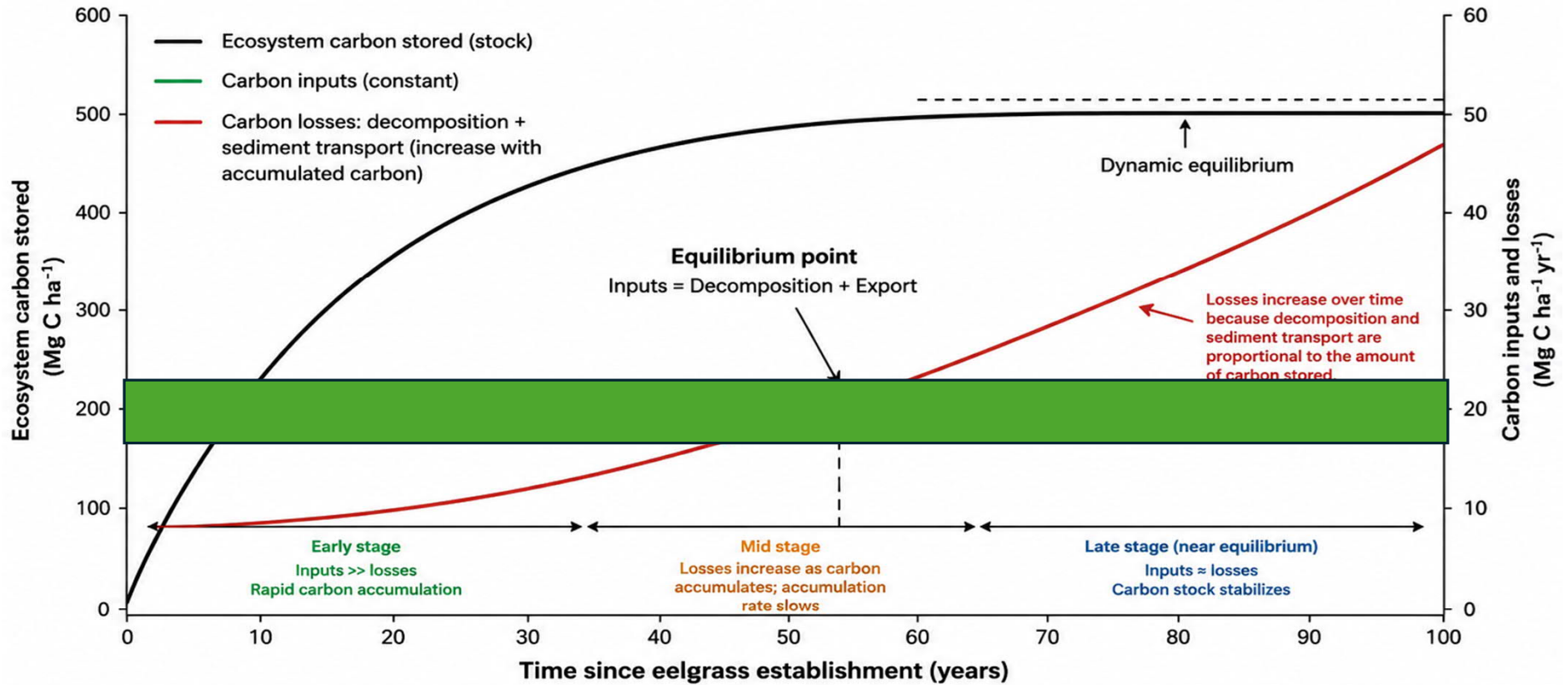
### Carbon losses (increase with carbon stored)

- Decomposition of organic matter
- Sediment transport / export (resuspension, burial below active layer, or advection)



*Conceptual model: As more carbon is stored in sediments, decomposition and sediment transport increase proportionally, reducing net accumulation until a dynamic equilibrium is reached.*

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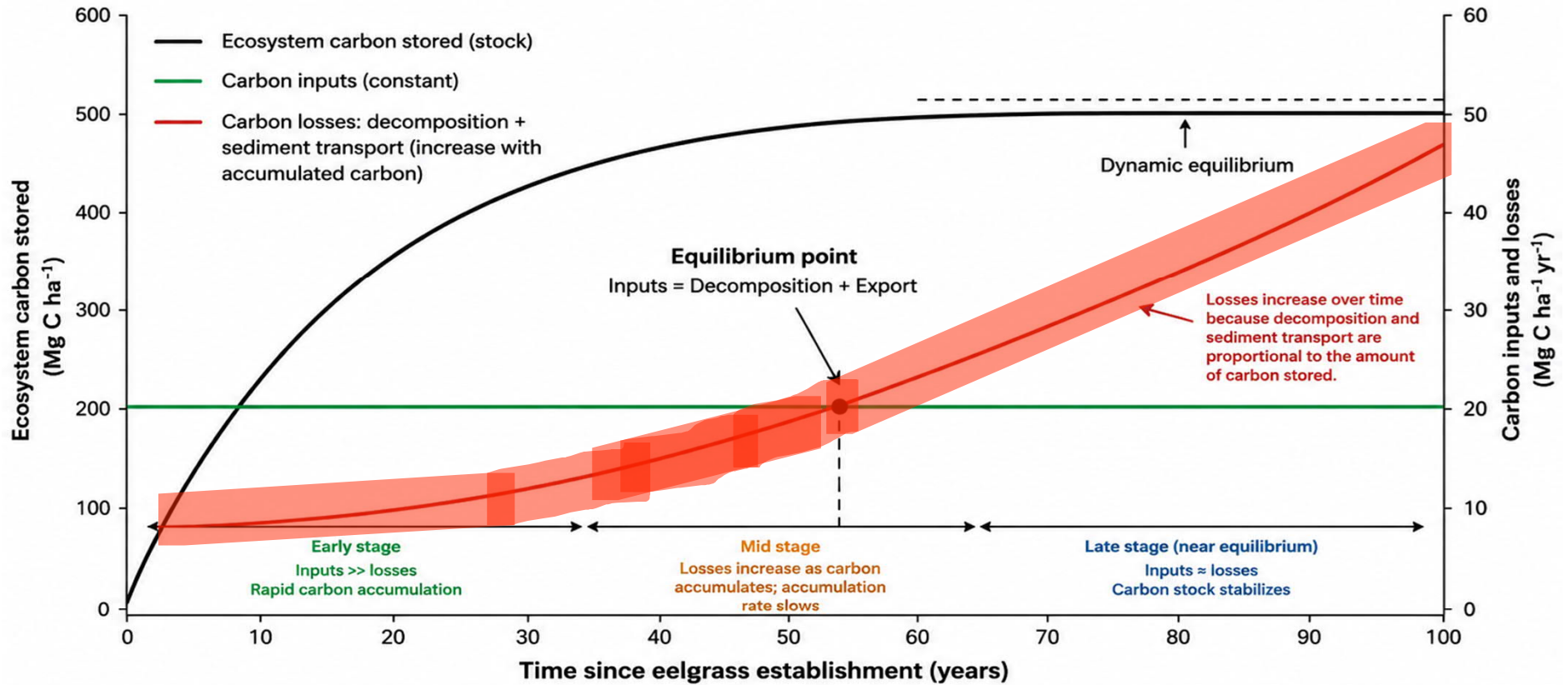
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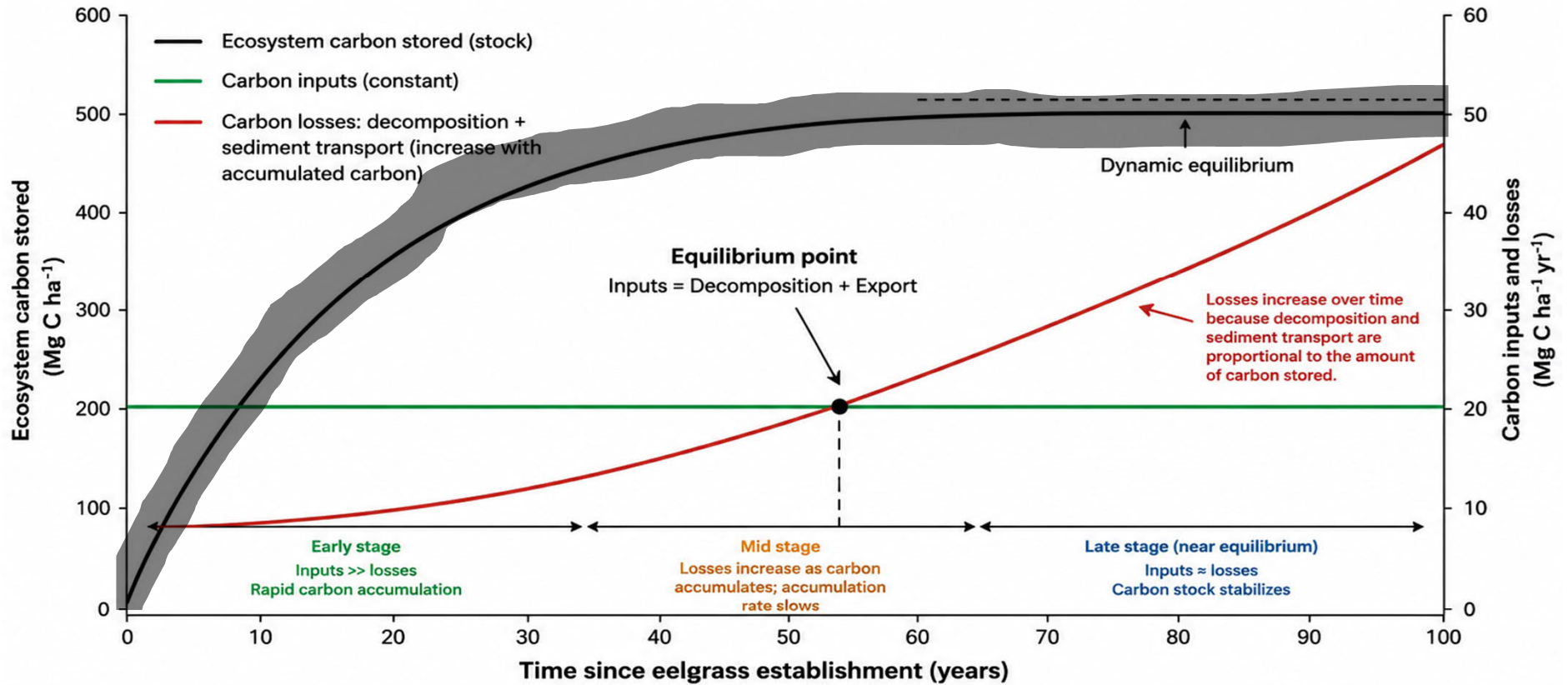
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# When Carbon Systems Are Stable vs Disturbed



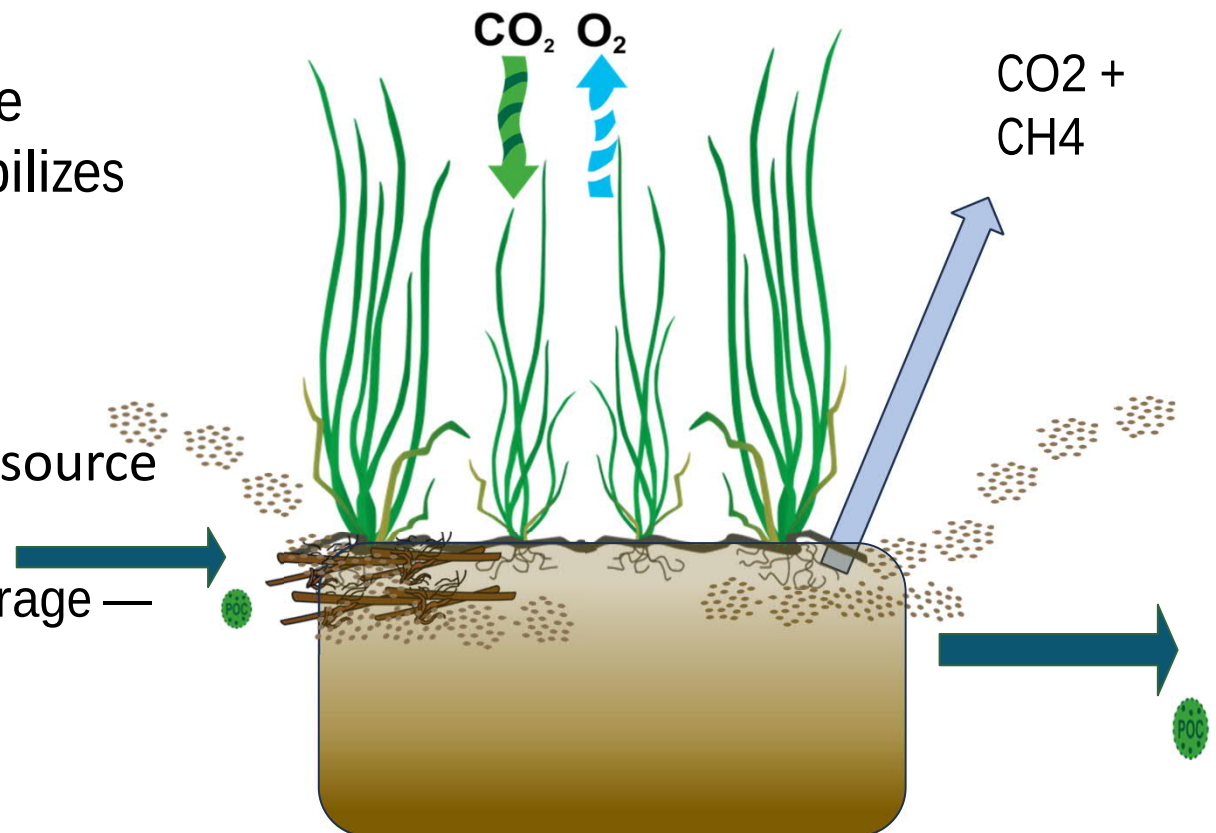
## Healthy systems

Carbon accumulates over time  
System balance gradually stabilizes

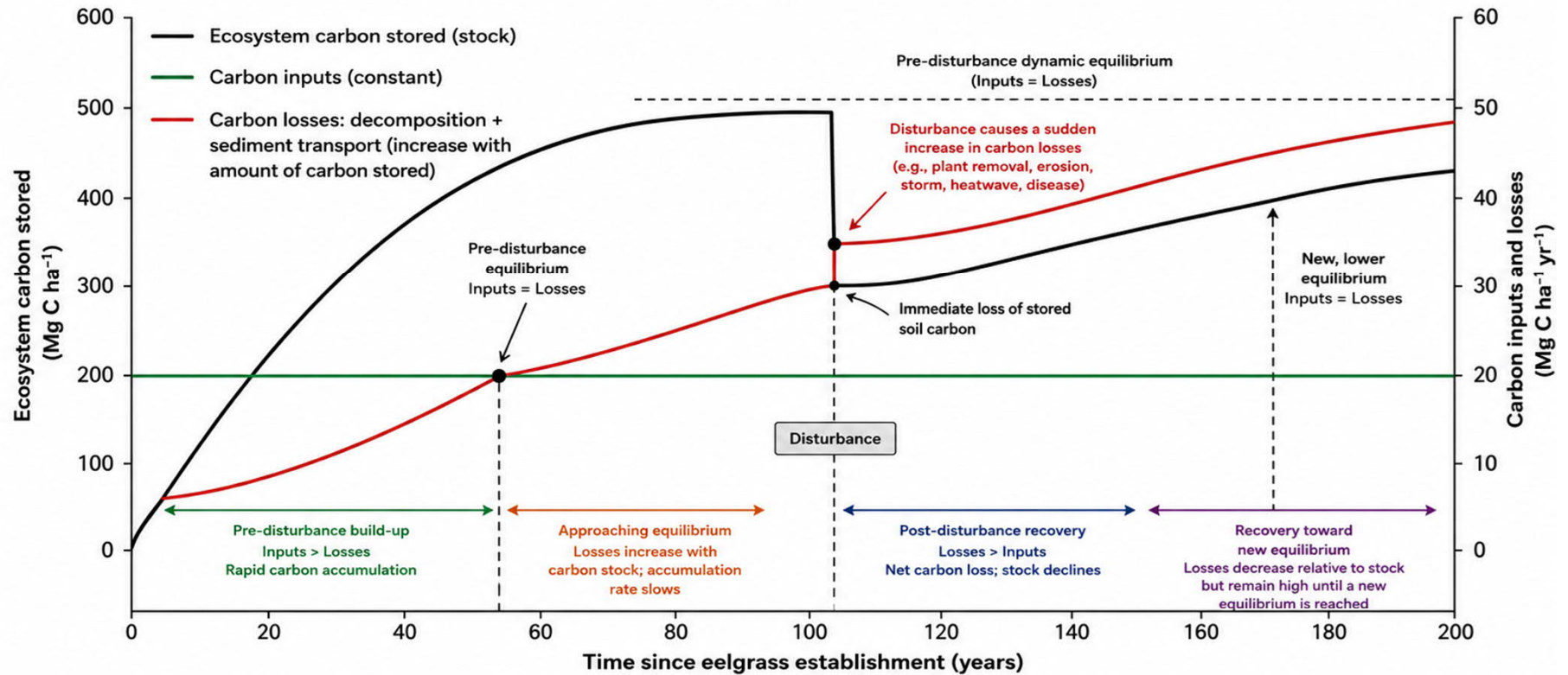
## Disturbance

Releases stored carbon  
Can shift system from sink  $\rightarrow$  source

Disturbance doesn't just stop storage —  
it can reverse it.

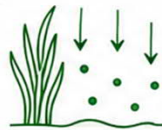


## Carbon accumulation in an eelgrass ecosystem over time with a disturbance



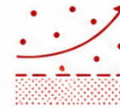
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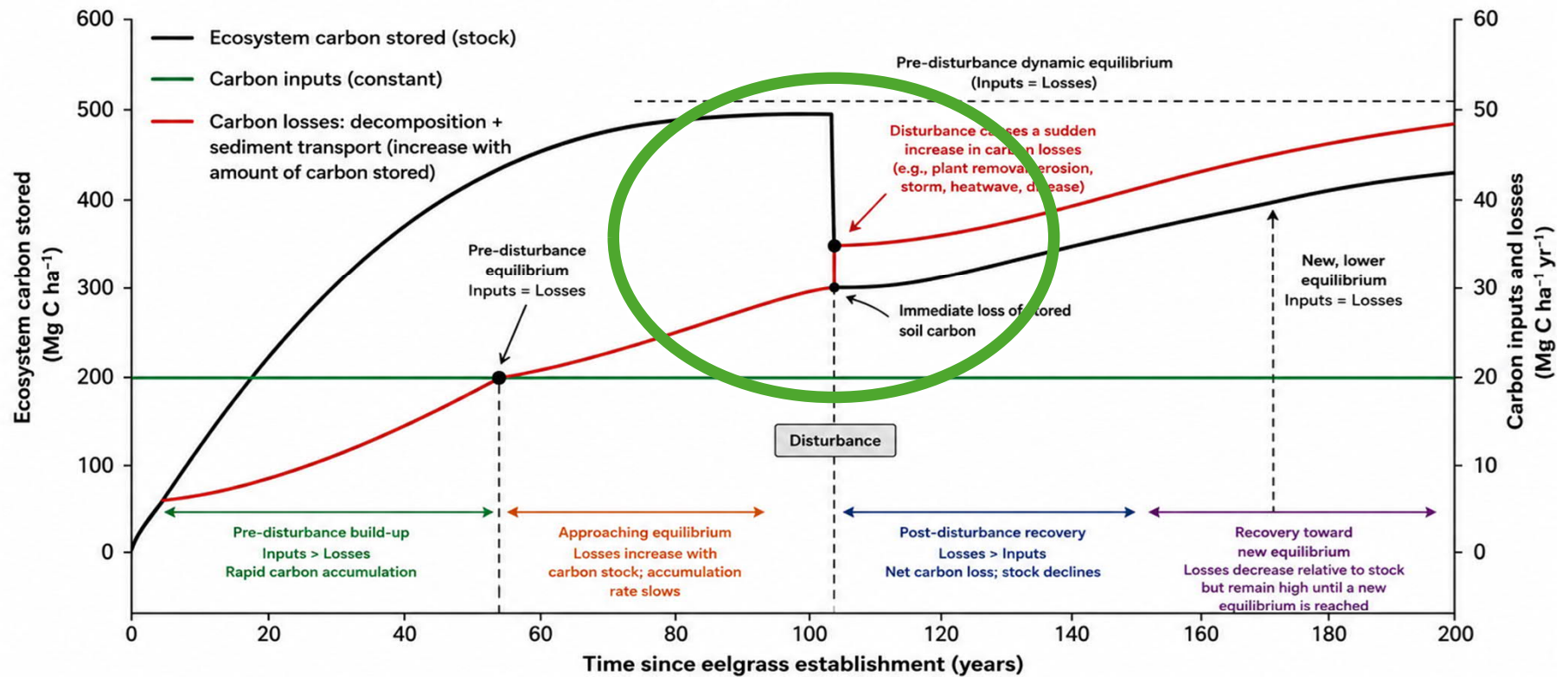
### Disturbance examples

- Storm / wave energy
- Physical disturbance (e.g., anchoring, dredging, trawling)
- Heatwave, disease, hypoxia
- Extreme erosion / sediment loss



*Conceptual model: Disturbance abruptly increases carbon losses and reduces stored carbon. Recovery is slow, with losses remaining elevated until a new, lower equilibrium with inputs is reached.*

## Carbon accumulation in an eelgrass ecosystem over time with a disturbance



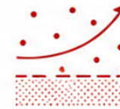
### Carbon inputs (constant through time)

- Carbon fixation through photosynthesis
- Allochthonous carbon deposition from water column



### Carbon losses (increase with carbon stored)

- Decomposition of organic matter
- Sediment transport / export (resuspension, burial below active layer, or advection)



### Disturbance examples

- Storm / wave energy
- Physical disturbance (e.g., anchoring, dredging, trawling)
- Heatwave, disease, hypoxia
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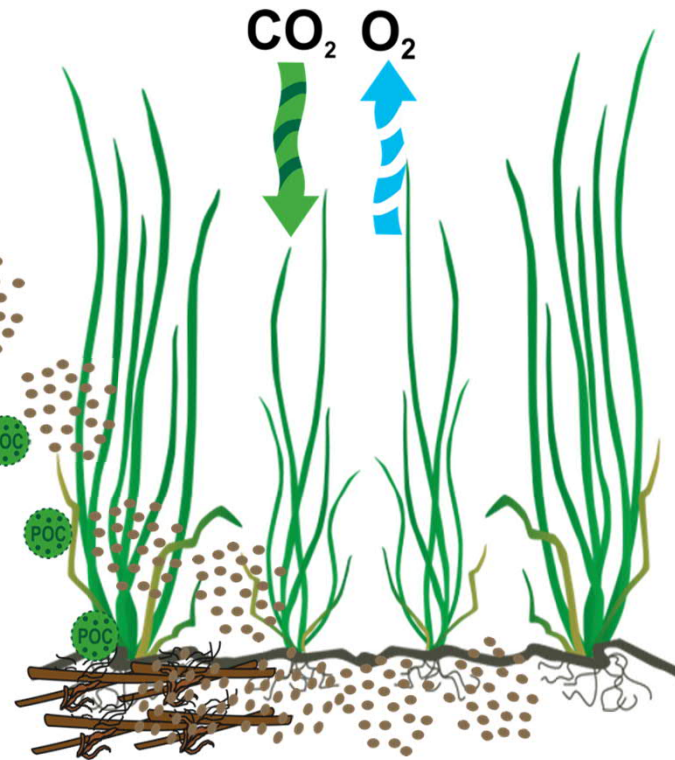
*Conceptual model: Disturbance abruptly increases carbon losses and reduces stored carbon. Recovery is slow, with losses remaining elevated until a new, lower equilibrium with inputs is reached.*

# Recipe for an effective blue carbon ecosystem

High rates of photosynthesis

Build up sediments through time

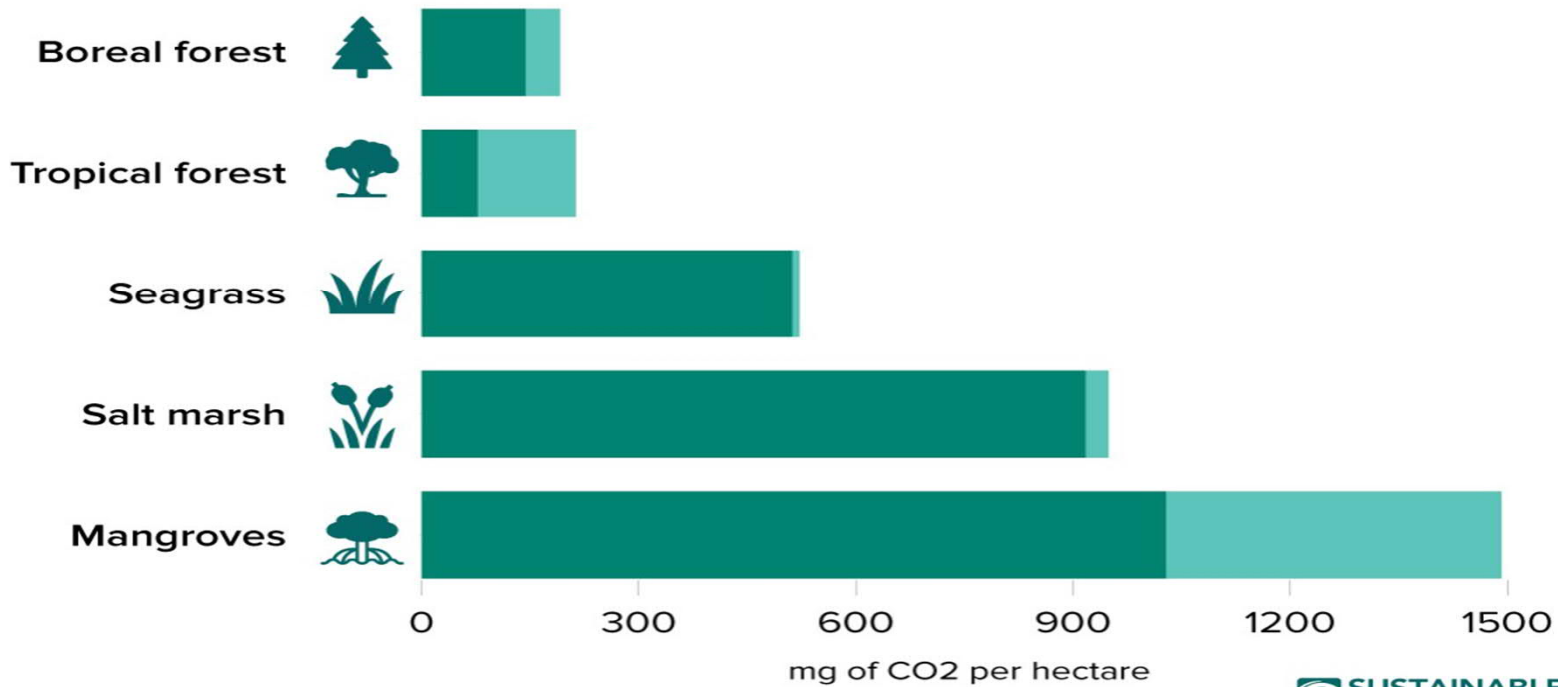
Capture carbon  
from surrounding  
ecosystems



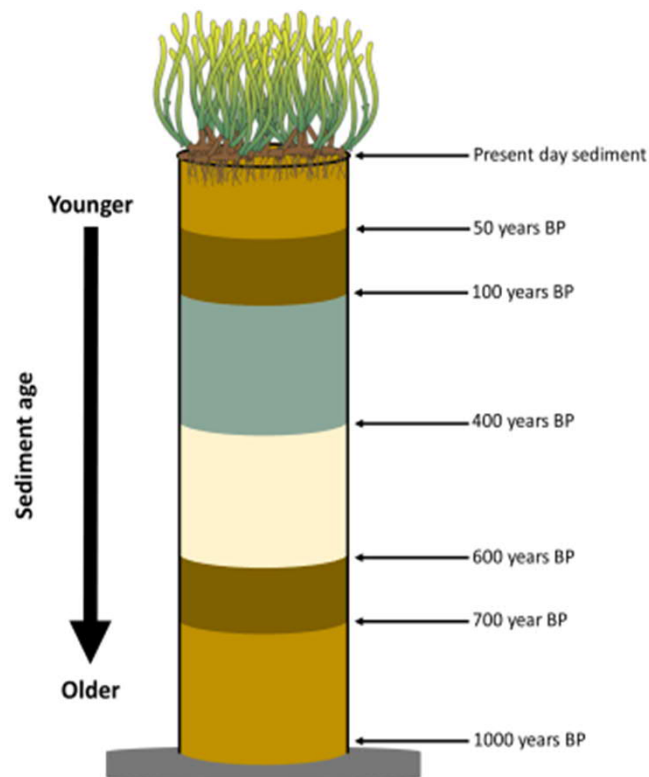
Low rates of  
decomposition

## Carbon Storage

soils biomass



(McLeod et al 2011)



## The carbon is in the soil

Far more carbon sits below ground in sediment than in the living leaves and roots above it.

## It stays locked away

Waterlogged, low-oxygen sediments slow decay, storing carbon for centuries to millennia.

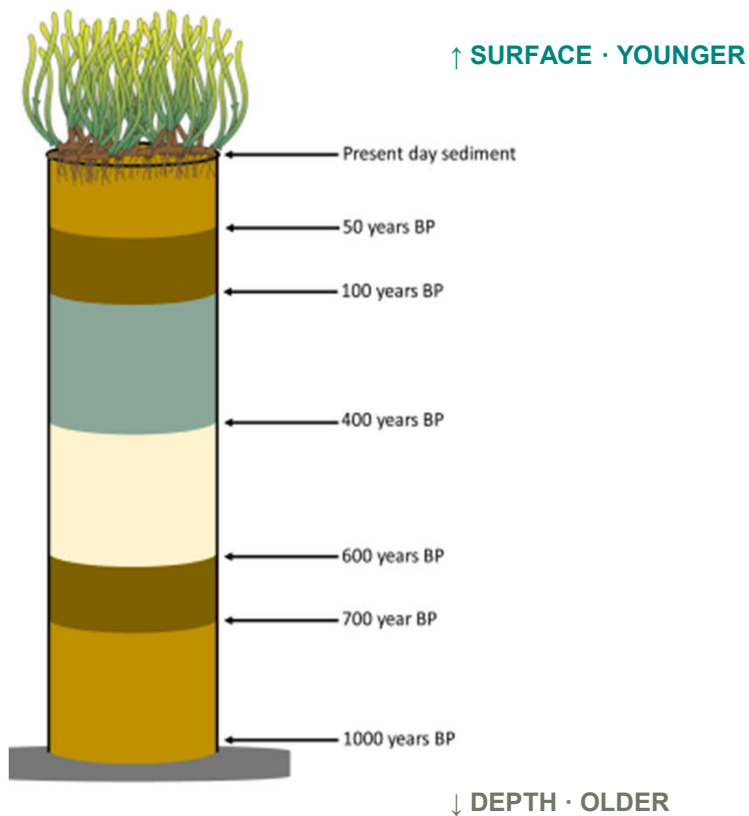
## Eelgrass carbon can be lost

When eelgrass is disturbed or destroyed, that stored carbon can be released again.

Carbon accumulates over long time periods — decades to centuries.

Disturbance can release decades–centuries of stored carbon rapidly.

# LAYERED TIME CAPSULES

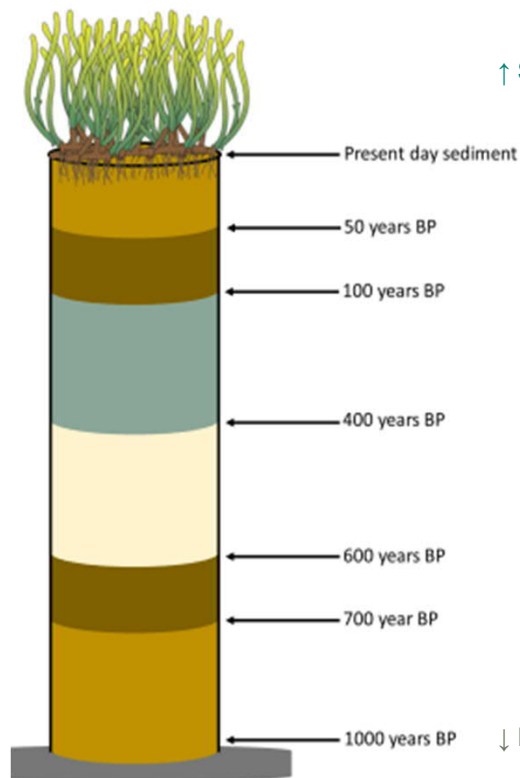


Pull a core out of the ground and you pull out a stack of time. The layers stay in order — **younger on top, older at the bottom.**

## Each layer records...

- **Carbon** — how much is stored at that depth
- **Accretion** — how fast sediment built up over time
- **History** — past disturbances, storms, and change

# LAYERED TIME CAPSULES



↑ SURFACE · YOUNGER

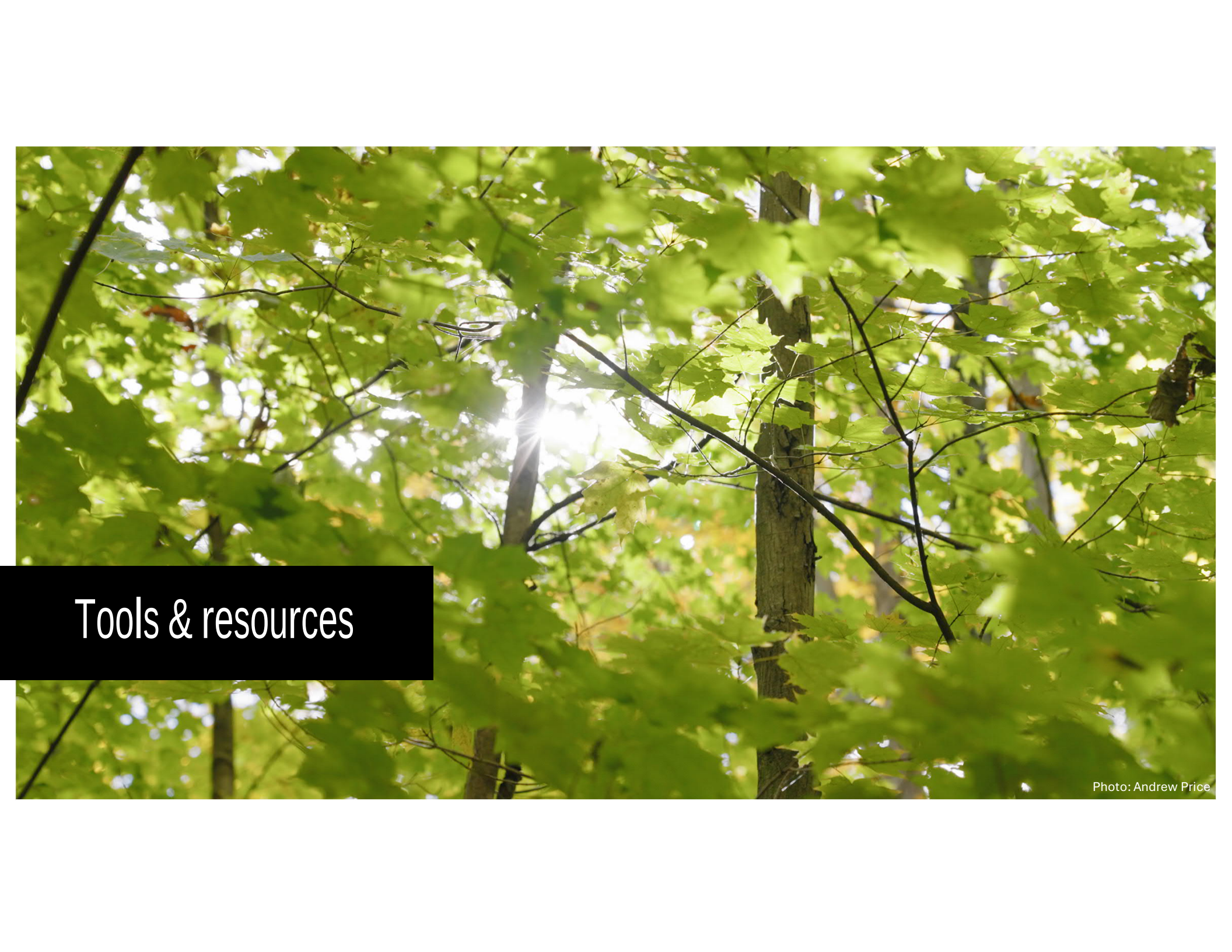
**Typically – higher % of carbon in top of soil, but soils are less dense**

- Recent plant
- Roots
- Less time for decomposition

**Soils typically have lower % Carbon, but soils are more dense**

- More time for decomposition

↓ DEPTH · OLDER

A low-angle photograph looking up into a dense forest canopy. Sunlight filters through the vibrant green leaves, creating a bright, starburst effect in the center. Several tree trunks and branches are visible, framing the central light source. The overall mood is peaceful and natural.

## Tools & resources

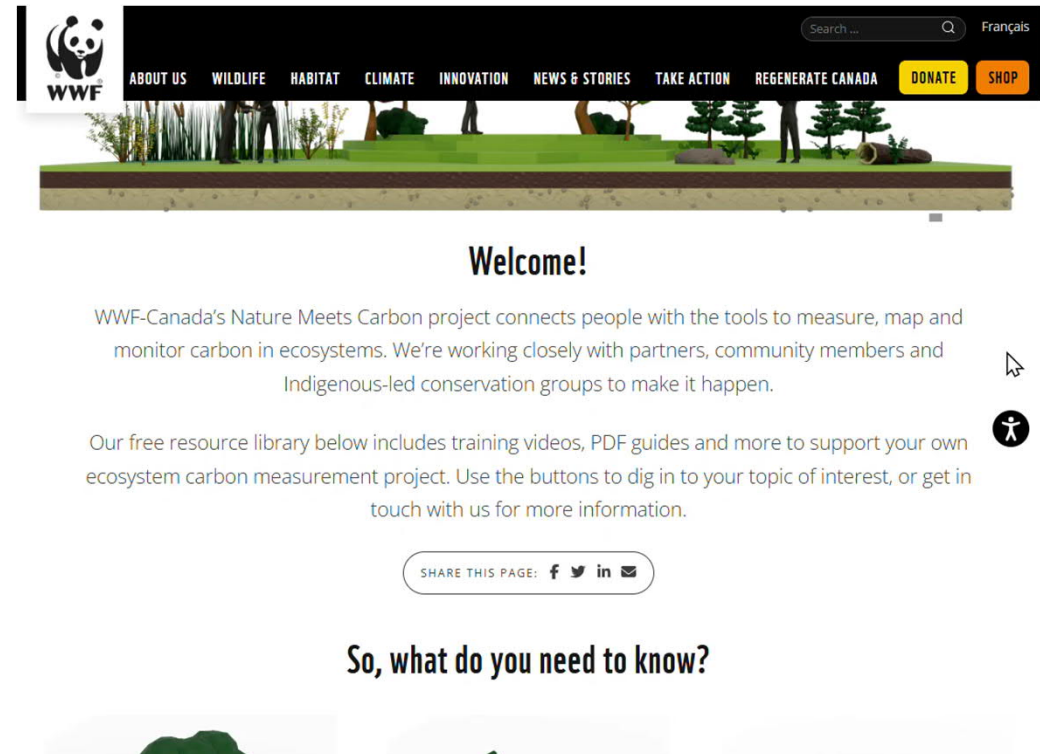
Photo: Andrew Price

# WWF Canada tools & resources

- WWF-Canada resources
  - PDF guides for peat and non-peat soils
  - Corresponding video explainers
  - Webinar training sessions
  - PDF sampling design guide CW1
  - Upcoming lab guide
  - Tools for determining number of plots and their locations

All available at:

<https://wwf.ca/carbon-measurement/>



CW1      **Speak to what this includes: determining the scale and scope of your carbon measurement project based on your budget, timeline, human resources, goals**

Clare Wark, 2024-09-30T17:48:04.608

CW1 0      **Alternatively there is time for a slide dedicated to this if we think it would be useful**

Clare Wark, 2024-09-30T17:54:42.405

# Other tools & resources

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## Drones

### Open-source products

- Open drone map
- Maps made easy
- Treetop and LidR

<https://tgoodbody.github.io/lidRtutorial/>

<https://github.com/tgoodbody/lidRtutorial>



lidRtutorial

CW1      **Should we include? I need to test still**

Clare Wark, 2024-09-30T17:54:11.324

CD1 0    **You can only test 2 trees before having to pay so its hard to say**

Cathal Doherty, 2024-10-10T18:39:22.836

CD2      **Should we add in the google sheet as well? Or we can demonstrate this after we have collected data.**

Cathal Doherty, 2024-10-02T13:29:44.955

CW2 0    **You mean the data sheets? Yep, I think that's a good idea.**

Clare Wark, 2024-10-02T14:33:16.468

CW3      **[@Cathal Doherty] bump drone tools/slides from end up to here?**

Clare Wark, 2026-05-22T18:09:08.666

CW3 0    **Added, feel free to edit/remove**

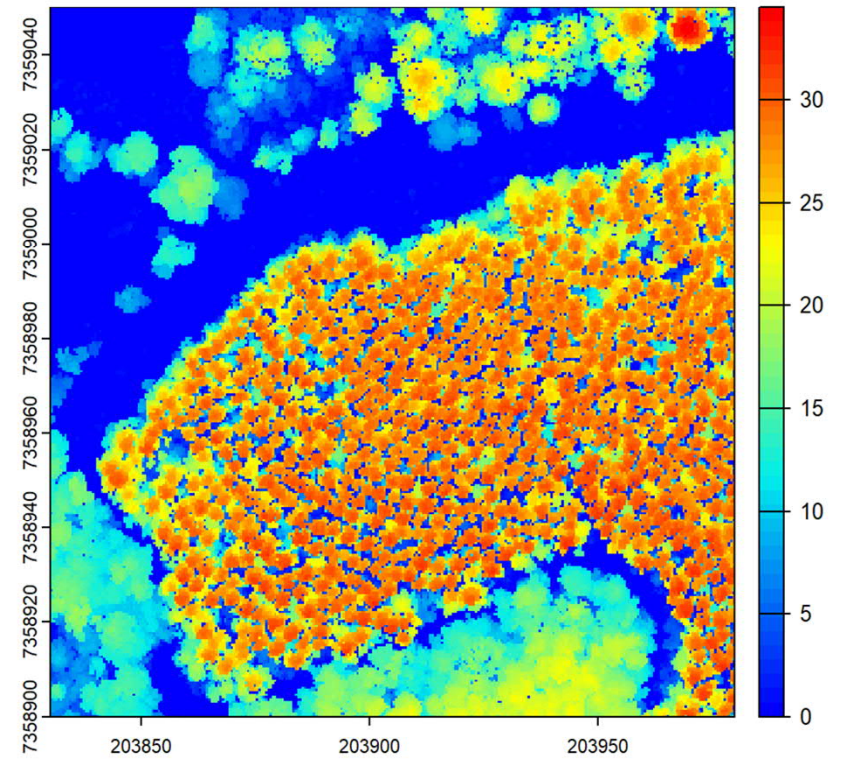
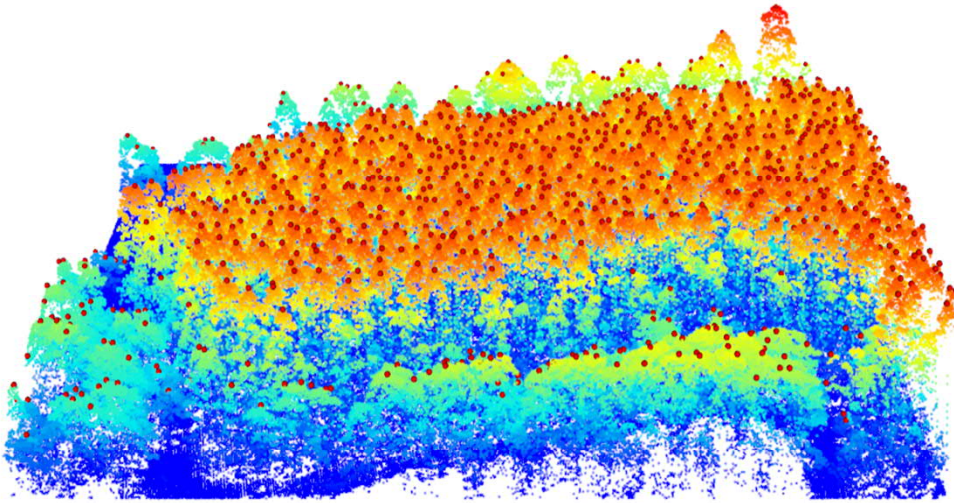
Clare Wark, 2026-05-22T19:09:03.103

CD3 1    **I hid these slides so they have them, but since were not covering trees I think we should skip**

Cathal Doherty, 2026-05-26T14:16:49.552

## Other tools & resources

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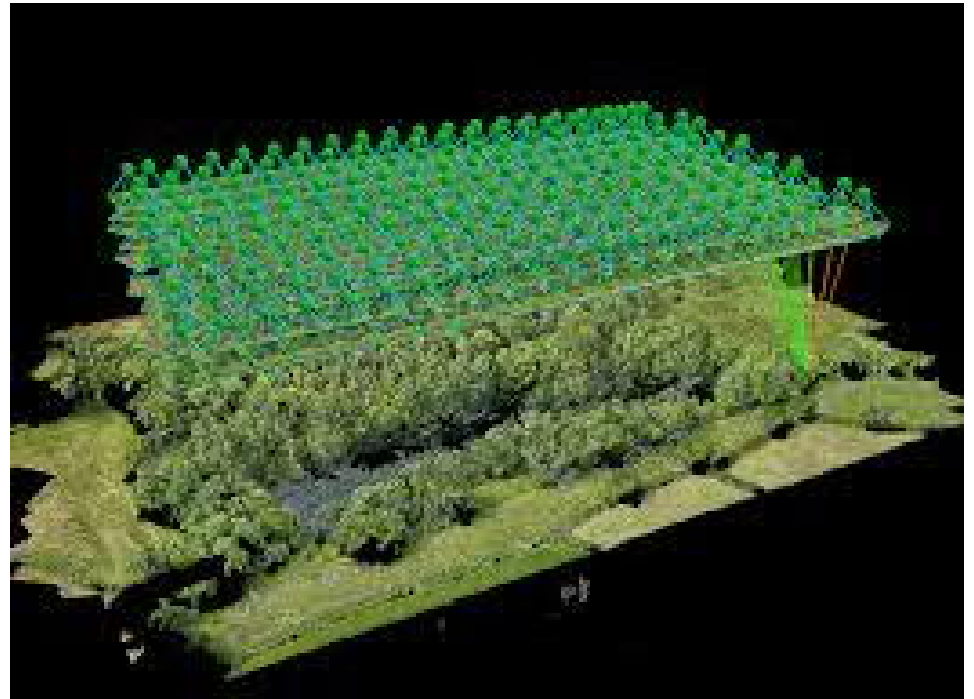
# Other tools & resources

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## Drones

### Commercial Products

- Pix4d
- Microsoft Agisoft
- Korotu Technology - DroneWatch





## Examples of carbon projects CW1

© Josh Stickles-Lohnes

CW1      **[@Cathal Doherty] can you review the three project examples I've included? I've used similar ones to what we included in the sampling design pdf but included a large one as well**

Clare Wark, 2026-05-01T18:49:32.934

CD1 0    **Thanks for adding, they look great. I'm going to add a couple extra details to show what each requires/outcomes**

Cathal Doherty, 2026-05-04T16:04:30.419

# Medium scale project example

First Nation Guardians with **100 km<sup>2</sup> forest and wetlands**

**GOAL** Carry out carbon measurement alongside other Guardian tasks and potentially use results to secure future funding

- Used Google Earth Engine to draw boundary
- Set up study sites based on ecosystem type, 10% acceptable error value
- Plots: 56 forest, 165 soils
- Strategy: systematic sampling
- Plot design: temporary 400 m<sup>2</sup> for trees & coring locations



# Medium scale project example

First Nation Guardians with **100 km<sup>2</sup> forest and wetlands**

**GOAL** Carry out carbon measurement alongside other Guardian tasks and potentially use results to secure future funding

## Outcomes:

- Carbon stock values & reporting statistics
  - 37.4 kgC/m<sup>2</sup> (+/- 10.6) in trees
  - 3,740 MtC (10% precision, 10% margin of error) in soils
- Carbon maps
- Answering more complex questions CW1

### Soil Carbon Statistics

Area: 6,987.5 ha  
Mean: 64.32 t/ha  
Std Dev: 13.50 t/ha  
CV: 21.0%  
Range: 40.39 - 118.26 t/ha  
 Using Manual CV: 67%

### Recommended Sample Size

**150 samples**

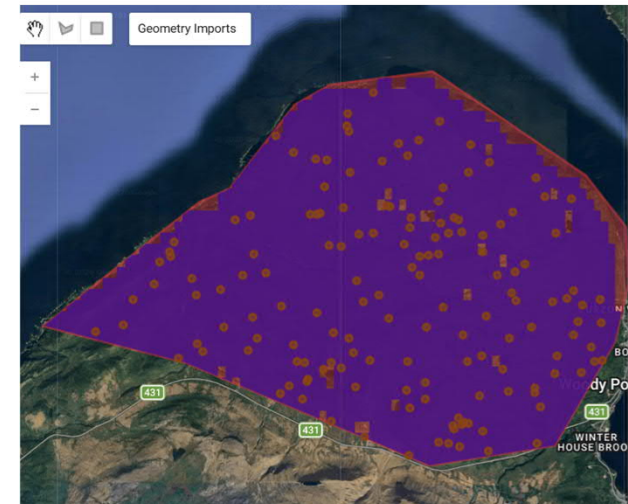
90% confidence, ±10% error

Margin of error: ±1.49 t/ha

Population (N): 2,794,995 | Sampling fraction: 0.01%

Apply to Points Field

✓ Applied 150 to points field



CW1      **[@Cathal Doherty]** - in this context how do these differ?  
Clare Wark, 2026-05-05T15:10:11.560

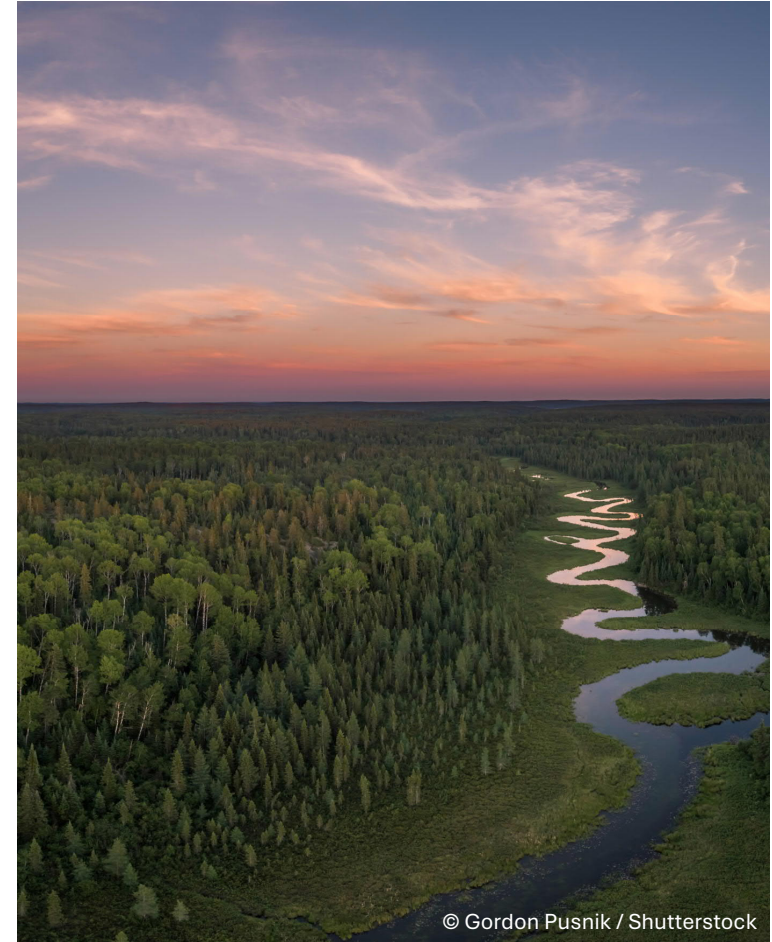
CD1 0    **They don't! I removed spatial outputs**  
Cathal Doherty, 2026-05-06T21:57:21.394

# Large scale project example

First Nation with **1,000 km<sup>2</sup> territory, encompassing forests and wetlands**

**GOAL** Assess carbon stored in newly acquired land prior to beginning sustainable forestry operations

- Used Google Earth Engine to draw boundary
- Set up study sites based on landcover, 10% acceptable error value
- Plots: 461 forest, 1,402 soil
- Strategy: stratified-random sampling
- Plot design: mix of temporary and permanent, 400 m<sup>2</sup> for trees & coring locations



© Gordon Pusnik / Shutterstock

# Large scale project example

First Nation with **1,000 km<sup>2</sup> territory, encompassing forests and wetlands**

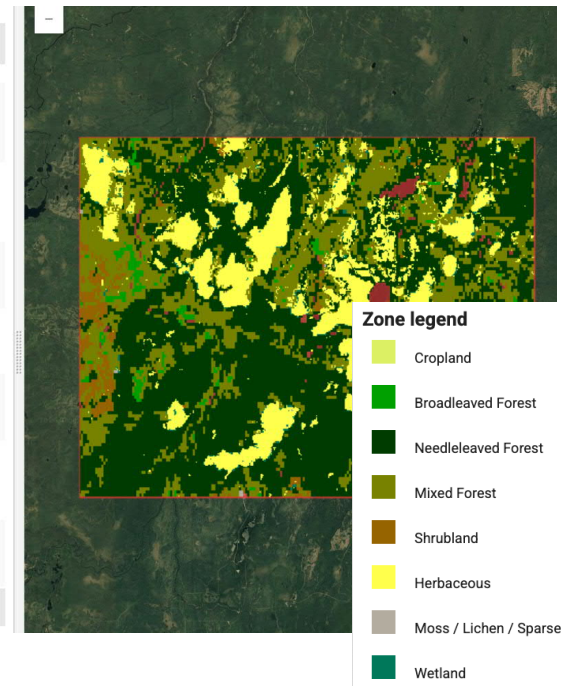
## GOAL

Assess carbon stored in newly acquired land prior to beginning sustainable forestry operations

### Outcomes:

- Carbon stocks & reporting statistics
- Carbon maps
- Answering complex questions
- Remote sensing and other tools to supplement and support work on the ground

| Plots per zone                                                                  |                        |          |       |       |      |
|---------------------------------------------------------------------------------|------------------------|----------|-------|-------|------|
| Include                                                                         | Zone                   | Area(ha) | %     | Prop. | Adj. |
| Prop. = proportional to area   Adj. = slightly reduced for larger zones (α=0.5) |                        |          |       |       |      |
| <input checked="" type="checkbox"/>                                             | Broadleaved Forest     | 3,393    | 1.4%  | 6     | 26   |
|                                                                                 | 41.7 t/ha              |          |       |       |      |
| <input checked="" type="checkbox"/>                                             | Needleleaved Forest    | 124,065  | 52.8% | 233   | 158  |
|                                                                                 | 44.8 t/ha              |          |       |       |      |
| <input checked="" type="checkbox"/>                                             | Mixed Forest           | 64,548   | 27.5% | 121   | 114  |
|                                                                                 | 43.2 t/ha              |          |       |       |      |
| <input checked="" type="checkbox"/>                                             | Shrubland              | 5,670    | 2.4%  | 10    | 34   |
|                                                                                 | 55.2 t/ha              |          |       |       |      |
| <input checked="" type="checkbox"/>                                             | Herbaceous             | 34,740   | 14.8% | 65    | 84   |
|                                                                                 | 71.6 t/ha              |          |       |       |      |
| <input checked="" type="checkbox"/>                                             | Moss / Lichen / Sparse | 126      | 0.1%  | 5     | 5    |
|                                                                                 | 61.0 t/ha              |          |       |       |      |
| <input checked="" type="checkbox"/>                                             | Wetland                | 2,259    | 1.0%  | 5     | 21   |
|                                                                                 | 57.4 t/ha              |          |       |       |      |
| Total                                                                           |                        |          |       | 445   | 442  |



# Desktop analysis of carbon



Photo: Casa Di Media Productions

CW1

**Speaker: [@Cathal Doherty]**

Clare Wark, 2026-05-01T13:56:53.658

# What is a desktop analysis?

---

- A method to estimate carbon in a specific area by using an online platform paired with carbon data



# Why start with a desktop analysis?

- Completing a desktop analysis of carbon doesn't require specialized skills or lots of time
- It can be used to determine where future carbon fieldwork should take place and if these areas overlap with other areas of interest



Photo: Clare Wark

# Overview of a desktop analysis

- To complete the carbon stock assessment, you will need the following:
  - Google Earth Engine account (free) or ArcGIS (paid platform)
  - If you're using ArcGIS you will need to download the raw carbon stock data (free)



Photo: Andrew Price

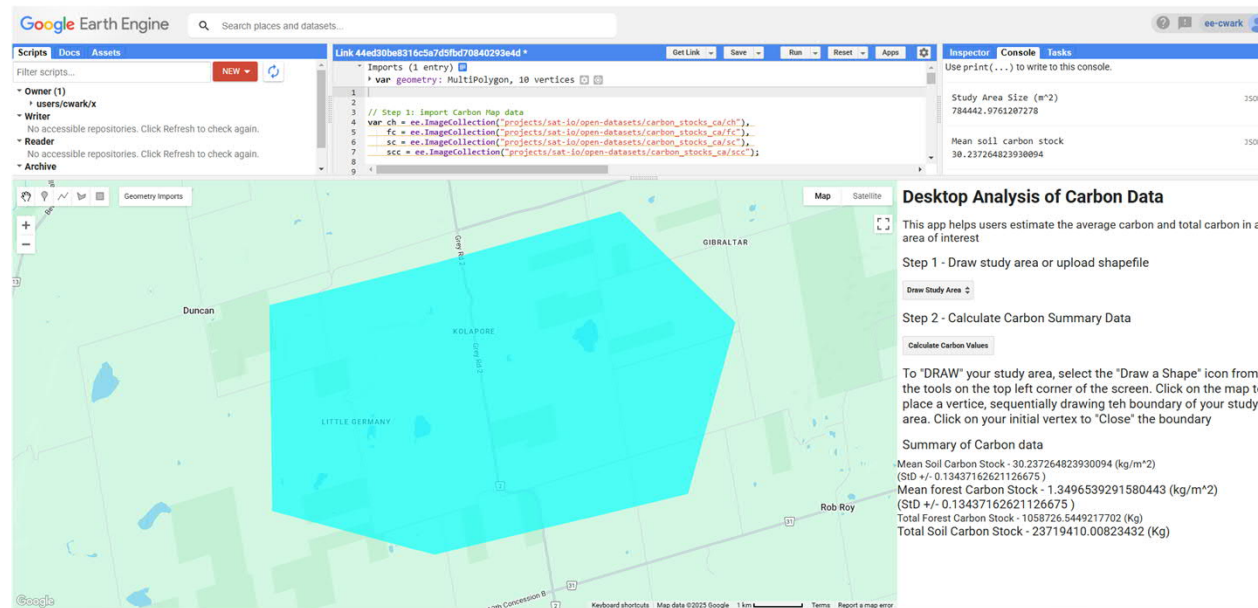
CW1

After this slide, begin screen sharing walk-through of process. GEE only? Or GEE and ArcGIS?

Clare Wark, 2025-01-24T17:41:44.388

# Steps to estimate carbon in Google Earth Engine (GEE) CW1

1. Open the GEE platform, login, and open the [carbon data app](#)
2. Upload your area of interest as a shapefile **or** draw your area as a polygon on the map
3. Click "Calculate Carbon Values" on right pane to get estimates for mean and total soil and forest carbon stocks
4. Save the file or click "Get Link" to share with other GEE users



GEE carbon data app

CW1      **Walk through 1 example on GEE and 1 on arcGIS**

Clare Wark, 2025-02-03T21:16:00.161

CW2      **No need to read through these steps, meant to serve as a reminder or point for people to pause at in recording**

Clare Wark, 2025-02-03T21:22:40.771

# Steps to estimate carbon in ArcGIS

---

1. Download the carbon map from WWF-Canada's [4TU.Researchdata](#) account and add to a new project in ArcGIS
2. Have your area of interest prepared in a shapefile or raster format and loaded in ArcGIS
3. Under the "Analysis" tab, click "Tools" to open the Geoprocessing pane
4. You can then search for "Zonal statistics as table"
5. Under "Input raster or feature zone data" select your data file selected area
6. Under "Input value raster" select the uploaded carbon map
7. Under "Output table" select where you would like the final output stored
8. Click "Run"
9. In the left pane, right click on your output table under the "Standalone Tables" tab and select "Open"
10. This table will show you the estimated mean of all terrestrial carbon in  $\text{kg/m}^2$  for your selected area
11. Multiply this value by the "Area" value to obtain an estimate for total terrestrial carbon kg for your selected area

- CW1      No need to read through these steps, meant to serve as a reminder or point for people to pause at in recording  
Clare Wark, 2025-02-03T21:22:50.864
- CW2      [**@Cathal Doherty**] after this slide we can switch to screen share and get people to take out their laptops and try their own desktop analysis on GEE  
Clare Wark, 2026-05-01T19:13:09.254

# Desktop Analysis Example

---

- Let's try running a desktop analysis
1. Living planet data hub - <https://datahub.wwf.ca/map/global-canada/carbon>
  2. GEE Desktop App - <https://cathalpdohertytestfolder1.users.earthengine.app/view/carbondesktopanalysis>
  3. GEE Planning app - <https://ee-cathalpdoherty2.projects.earthengine.app/view/stratification-sampling-tool>

CD1

Cal to add links and share screen

Cathal Doherty, 2026-05-05T13:16:58.015

# After a desktop analysis

---

- In some scenarios, desktop analysis may not be adequate for the work you are hoping to complete
- For example:
  - Estimating carbon in an area where there is very limited field data
  - Estimating carbon in a forest that was recently affected by a severe weather event
  - If you are interested in carbon estimates in soils deeper than 1m
  - If you are interested in estimating carbon in coastal areas
  - If you want to develop a carbon crediting or offset program



Photo: Clare Wark

## After a desktop analysis

- Desktop analysis is only the first step
  - What comes next is the collection of data through field measurements
- Field measurements for carbon can be taken for trees, non-tree vegetation, soils, and sediments



# Sampling design

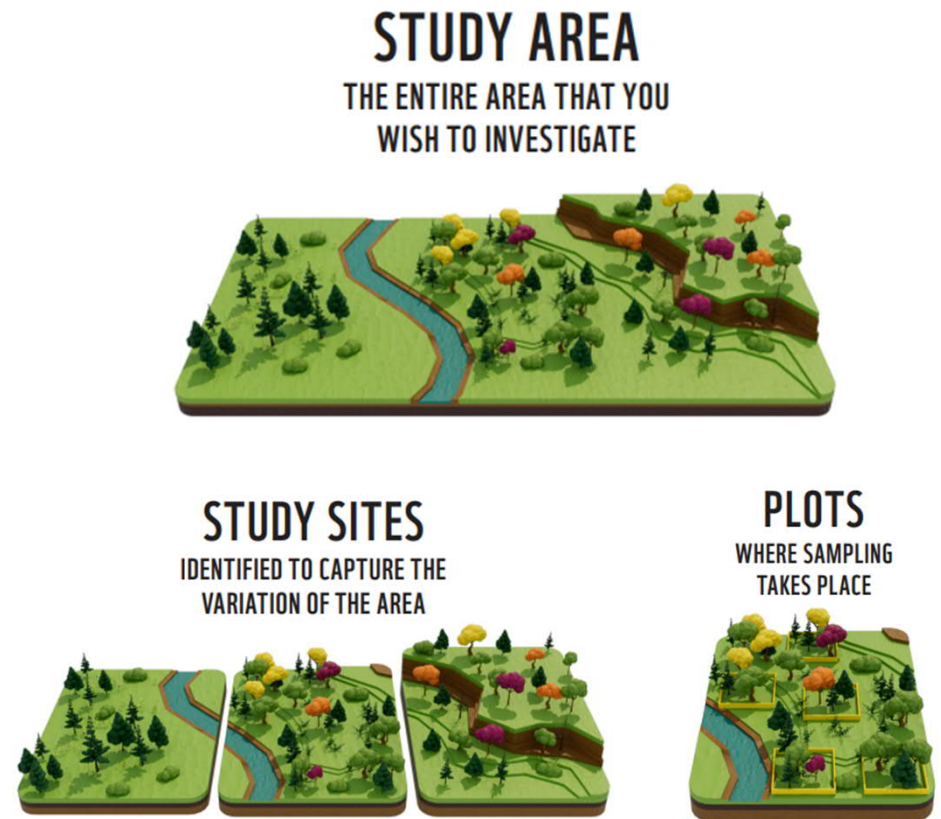


# What is a sampling design? CW1

Connects field work to intended outcomes

How?

- Provides a framework for choosing what and where to sample to estimate the carbon stored in an area
- Sampling designs allow for the strategic measurement of plots and sites within a larger study area
- Study sites are identified to capture variation of the area
- Plots are laid out within sites to obtain a representative sample of the study site



CW1      **[@Cathal Doherty]** animation added, have the room guess before reading through slide  
Clare Wark, 2026-05-05T15:18:07.227

# What is a sampling strategy?

- A sampling strategy is a plan for deciding where sites and plots will be located within the study area
- A sampling strategy maps out the plots and study sites to most effectively estimate the carbon stock for the larger study area
- Each type of sampling strategy has benefits and limitations

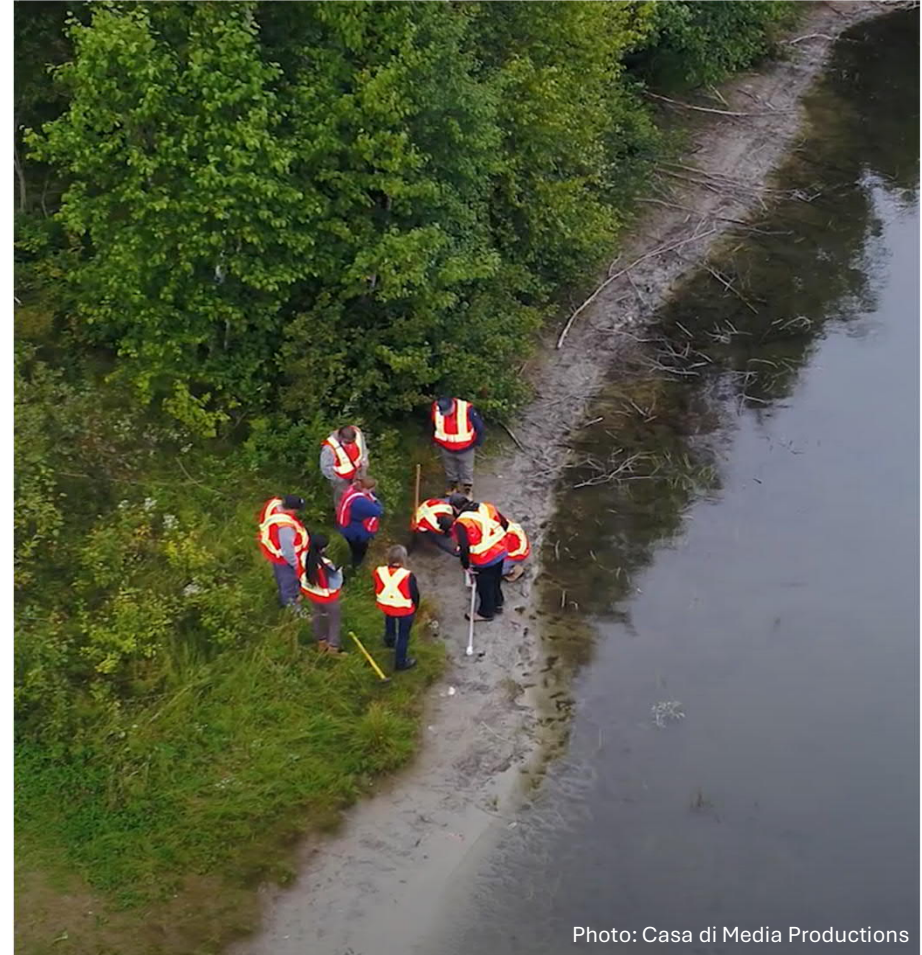


Photo: Casa di Media Productions

# Sampling strategy comparison

## CONVENIENCE/PRACTICAL SAMPLING



Plots are selected where sampling is most accessible.

## RANDOM SAMPLING



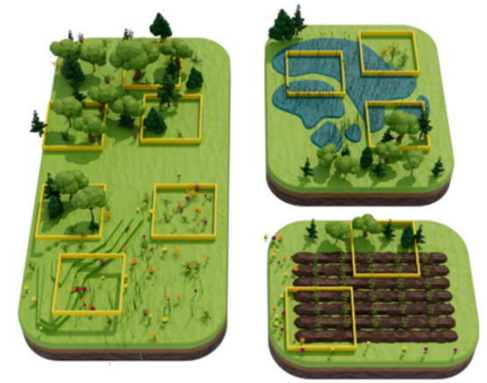
Plots are randomly selected across the study area.

## SYSTEMATIC SAMPLING



Plots are selected at regular intervals across the study area.

## STRATIFIED-RANDOM SAMPLING



Plots are randomly selected **after** the study area is divided into study sites based on certain characteristics.

# Determining the right approach for your sampling design

---

- **Level 1: Basic**
  - Great for community or pilot project
  - Requires minimal resources
  - Data is not detailed enough for formal reporting
- **Level 2: Intermediate**
  - Higher accuracy
  - Ideal for reporting purposes
  - Requires more planning and support
- **Level 3: Advanced**
  - Highest accuracy
  - Necessary for MRV reporting and carbon crediting programs

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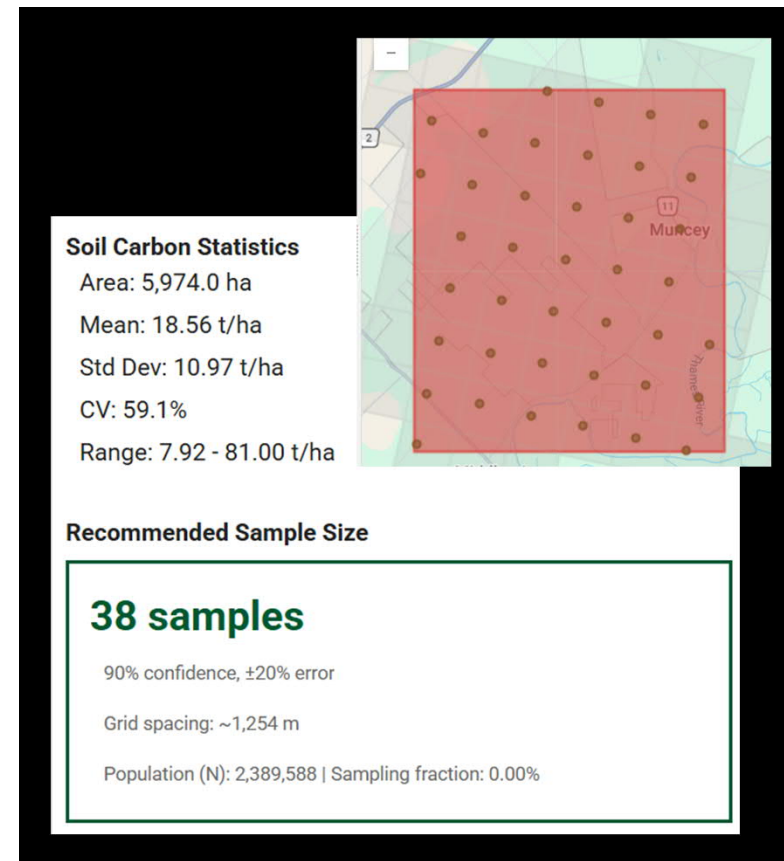
|                                                       |             |
|-------------------------------------------------------|-------------|
| <b>User inputs:</b>                                   |             |
| a) Size of total study area (m <sup>2</sup> )         | 1900        |
| b) Margin of error (acceptable range)                 | 5           |
| c) Confidence level (level of precision; alpha value) | 0.1         |
| Default Values - Can be updated by user:              |             |
| Forest Carbon Mean Value (from wwf-canada carbon map) | 4.14        |
| Forest Carbon SD Value (from wwf-canada carbon map)   | 1.8         |
| Soil Carbon Mean Value (wwf-canada carbon map)        | 13.2        |
| Soil Carbon SD Value (wwf-canada carbon map)          | 10          |
| Z-statistic (soils)                                   | 1.281551566 |
| Z-statistic (forests)                                 | 1.281551566 |
| <b>Output:</b>                                        |             |
| Number of plots (n)                                   | 4.119705796 |

## Non-spatial outputs

- Carbon stock density (kg/m<sup>2</sup>)
- Compare to other areas/baselines

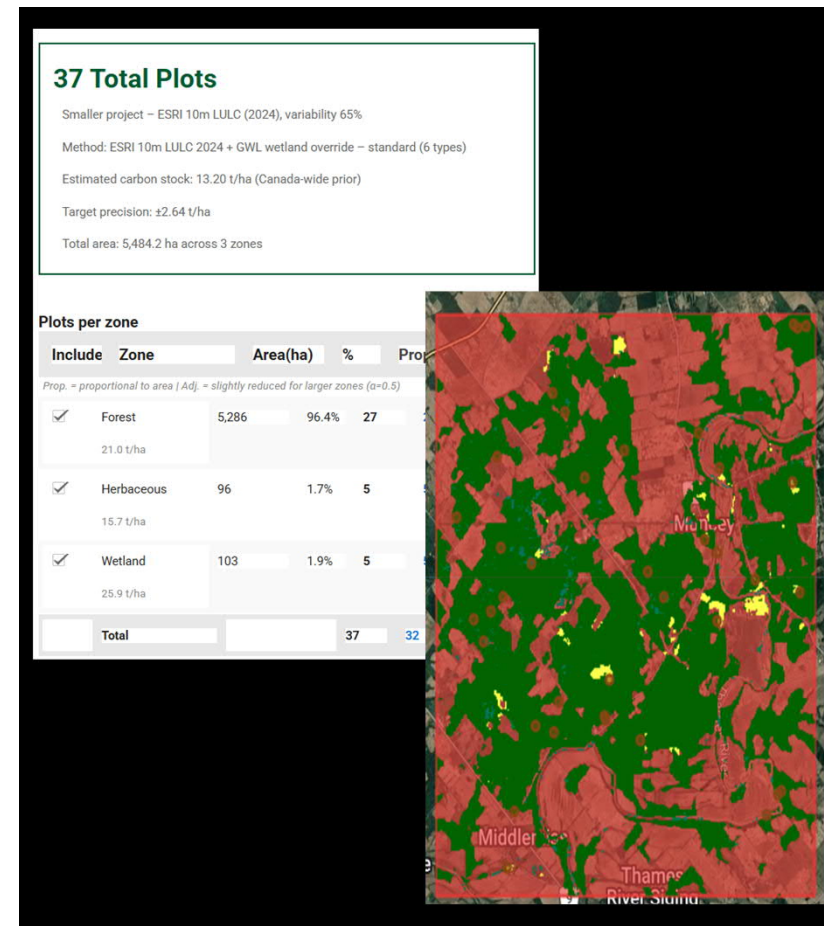
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# Applying a sampling design

1. Identify the boundary of the study area
2. Determine the sample allocation
3. Determine the sample distribution
4. (optional) Stratify study area
5. Select a plot design
6. Implement field measurement protocols
7. Results & assessment of data



TOTAL CARBON PER TREE

$$\left( \text{ABOVE-GROUND BIOMASS} + \text{BELOW-GROUND BIOMASS} \right) \times 0.5$$

| J14 |            |         |                             |                                                                                                                                         |
|-----|------------|---------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
|     | A          | B       | C                           | D                                                                                                                                       |
|     | study area | site id | site area (m <sup>2</sup> ) | Total carbon stock of study site (kg C) = Average carbon stock of study site (kg C/m <sup>2</sup> ) * Study site size (m <sup>2</sup> ) |
| 1   |            |         |                             |                                                                                                                                         |
| 2   | DB         | DB-01   | 100                         | 5461.5852557173                                                                                                                         |
| 3   | DB         | DB-02   | 100                         | 7424.818652                                                                                                                             |
| 4   | DB         | DB-03   | 100                         | 4160.851443                                                                                                                             |
| 5   | SUM        |         | 300                         | 17047.2553505744                                                                                                                        |

CD1      I changed the order of stratification from 2 to 4 to keep in line with the sampling plan we sent over  
Cathal Doherty, 2026-05-31T16:14:14.100

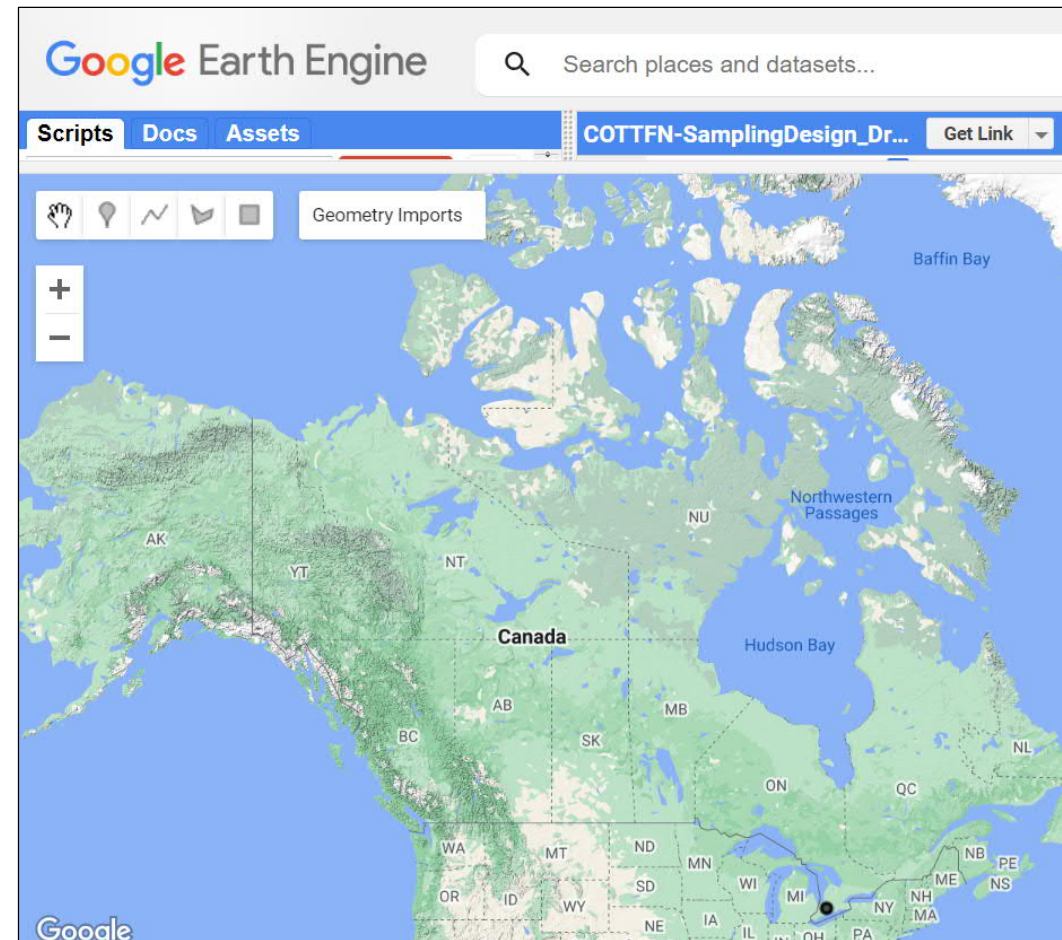
# Using Google Earth Engine (GEE)

## Why GEE?

- Free mapping software
- Many open-source tools for carbon monitoring
- Compatible with many types of geo data (shapefiles, tiff, CSV)
- Must have a google account

## Other open-source tools:

- QGIS
- Grass GIS
- <https://gisgeography.com/free-gis-software/>

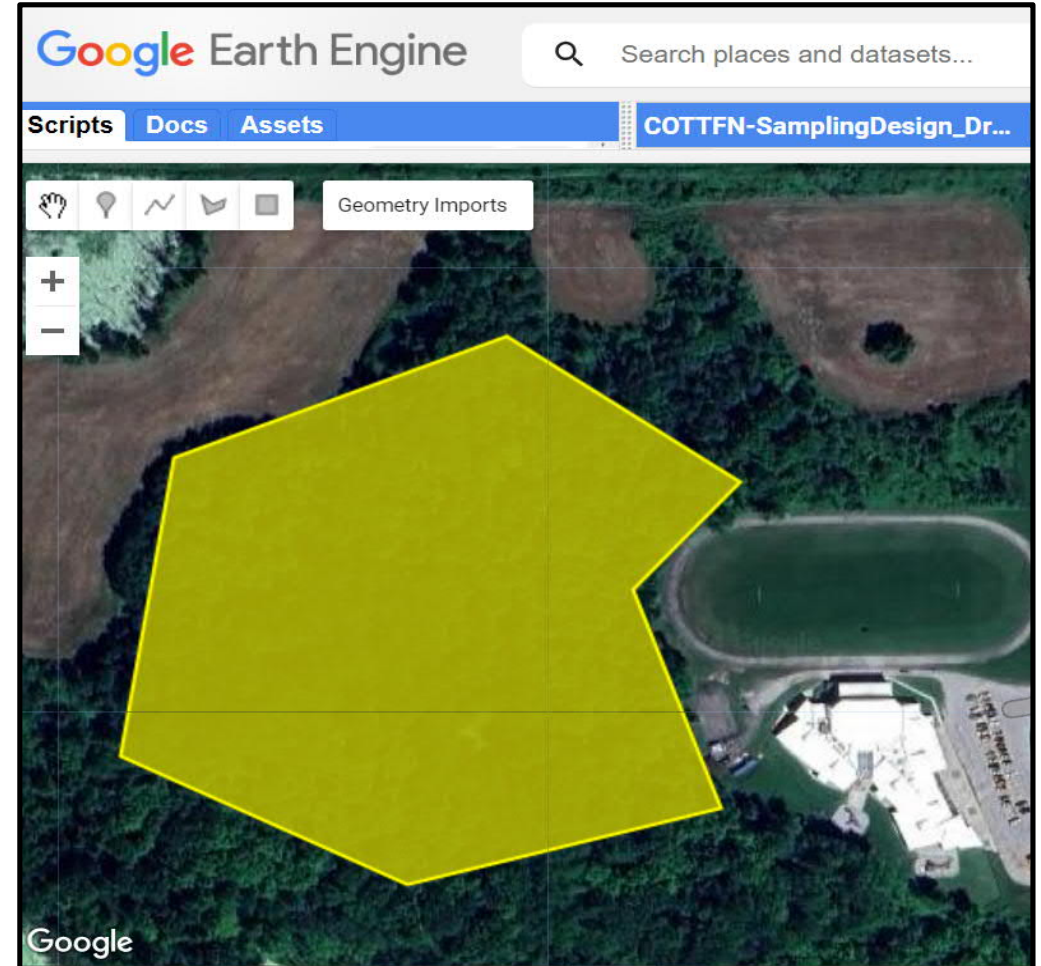


# Step 1: Identify the boundary of the study area

1

Two options for defining study area:

- A) Draw** your study area using a polygon/geometry function
- B) Import** an existing boundary layer
  - Shapefile
  - CSV format





CD1      Alderville uses ArcGIS so I'll remove these and replace with ESRI stuff  
Cathal Doherty, 2026-05-31T16:04:12.252

## Step 2: Sample Allocation

2

The **optimal** number of samples depends on:

A) The **size** of your study area

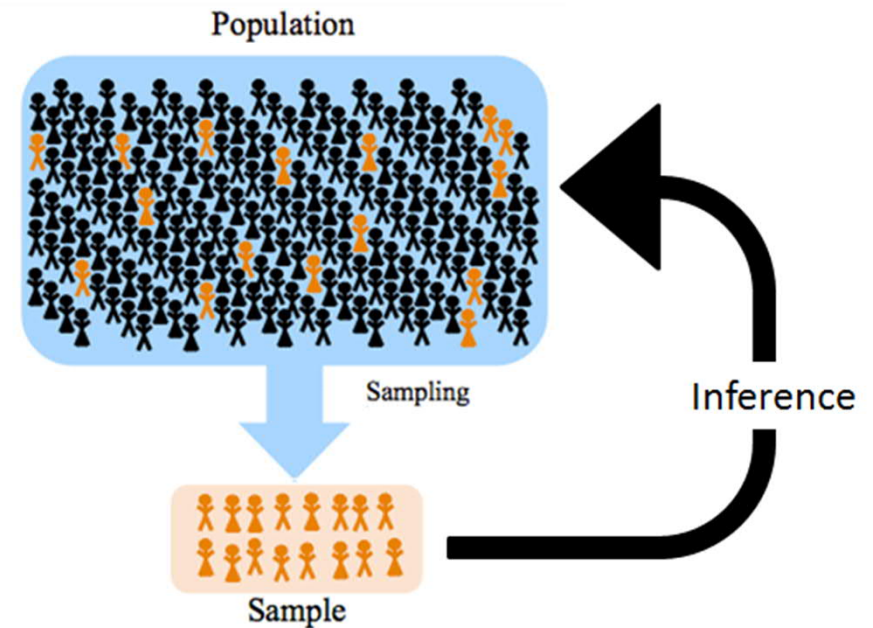
Larger study areas require more sites to accurately estimate the carbon stock compared to smaller study areas

B) What level of **accuracy** do you expect?

What is the probability that your measured values are within a defined level of precision

**Probability** (Alpha) = The percent chance your estimate is within your desired range

**Allowable error** = desired range (% +/-) of the average value you are measuring



## Step 2: Sample Allocation

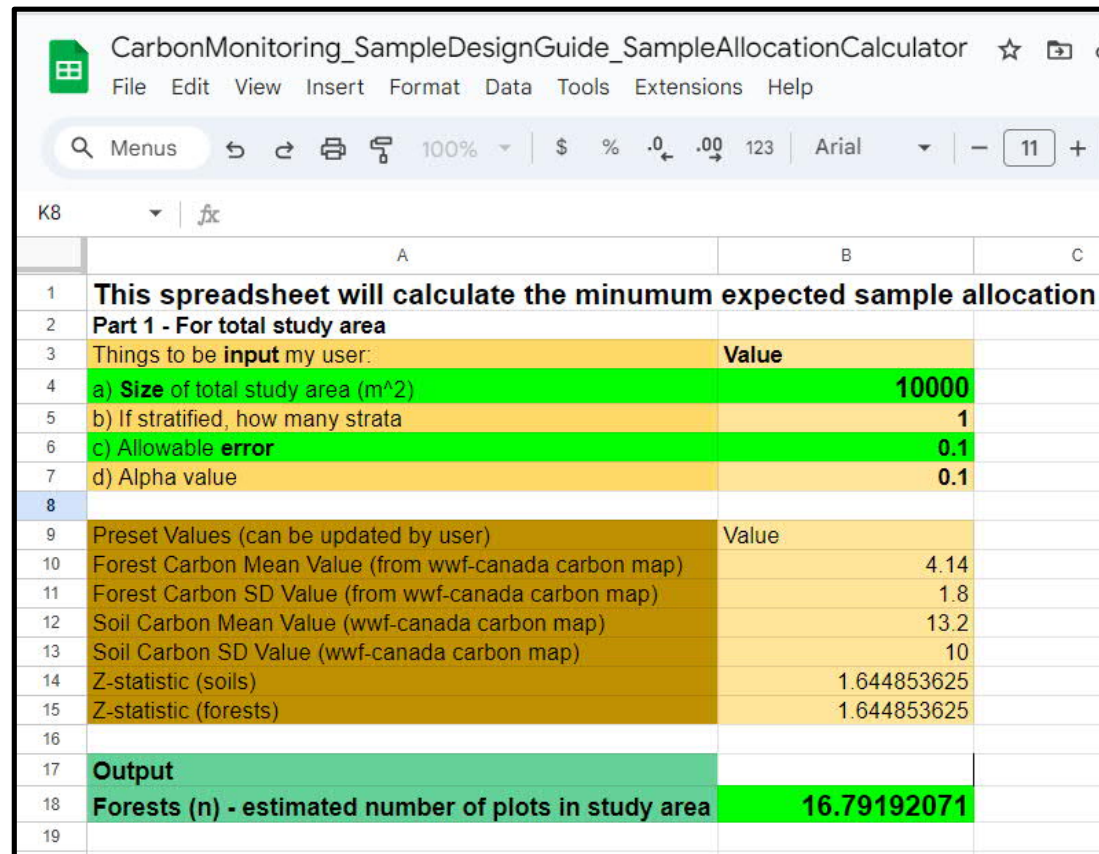
2

### Required Inputs:

- **Size** of study area (m<sup>2</sup>)
- **Allowable error** (default = 20%)
- Desired level of **precision** (default = 10%)

### Optional:

- Mean
- Standard deviation



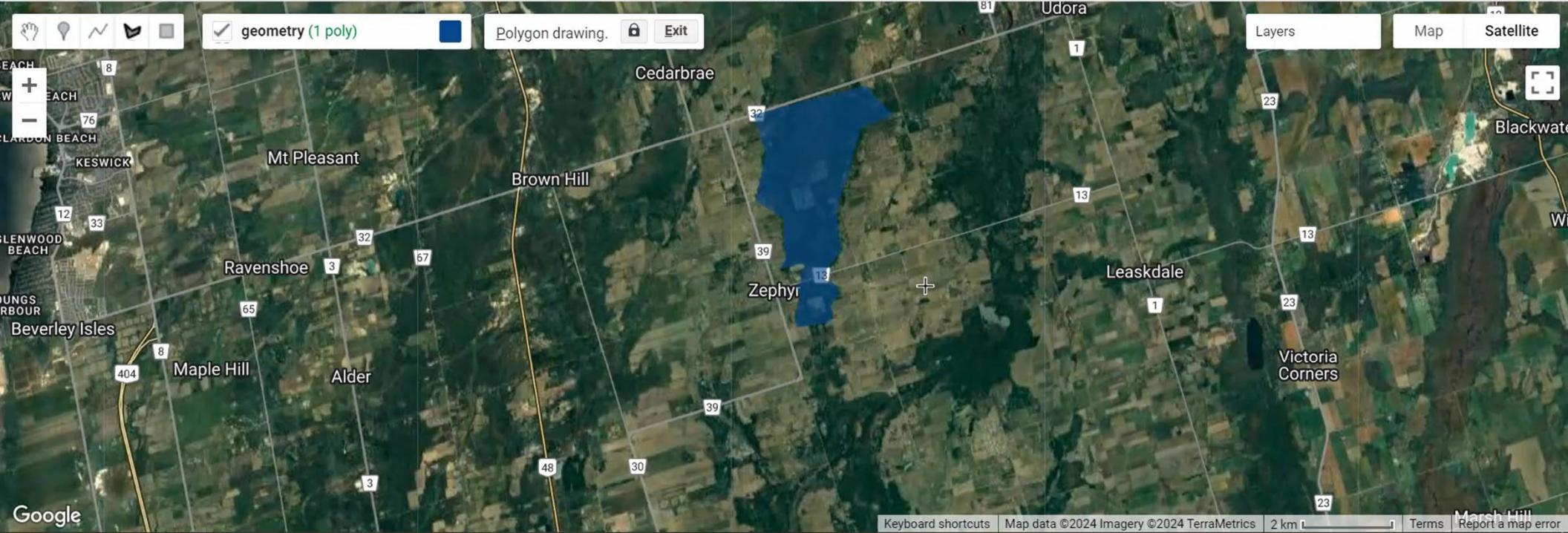
The screenshot shows a Google Spreadsheet with the following data:

|    | A                                                                             | B           | C |
|----|-------------------------------------------------------------------------------|-------------|---|
| 1  | <b>This spreadsheet will calculate the minimum expected sample allocation</b> |             |   |
| 2  | <b>Part 1 - For total study area</b>                                          |             |   |
| 3  | Things to be input my user:                                                   | Value       |   |
| 4  | a) Size of total study area (m <sup>2</sup> )                                 | 10000       |   |
| 5  | b) If stratified, how many strata                                             | 1           |   |
| 6  | c) Allowable error                                                            | 0.1         |   |
| 7  | d) Alpha value                                                                | 0.1         |   |
| 8  |                                                                               |             |   |
| 9  | Preset Values (can be updated by user)                                        | Value       |   |
| 10 | Forest Carbon Mean Value (from wwf-canada carbon map)                         | 4.14        |   |
| 11 | Forest Carbon SD Value (from wwf-canada carbon map)                           | 1.8         |   |
| 12 | Soil Carbon Mean Value (wwf-canada carbon map)                                | 13.2        |   |
| 13 | Soil Carbon SD Value (wwf-canada carbon map)                                  | 10          |   |
| 14 | Z-statistic (soils)                                                           | 1.644853625 |   |
| 15 | Z-statistic (forests)                                                         | 1.644853625 |   |
| 16 |                                                                               |             |   |
| 17 | <b>Output</b>                                                                 |             |   |
| 18 | Forests (n) - estimated number of plots in study area                         | 16.79192071 |   |
| 19 |                                                                               |             |   |

<https://docs.google.com/spreadsheets/d/1T7I1ydLIVUBzDSEH8gwdUZb6JqG9x5Orfpi0up4hdOQ/edit?usp=sharing>

Scripts Docs Assets Test1 (copy) \* Get Link Save Run Reset Apps Inspector Console Tasks

Filter scripts... NEW Imports (1 entry) Use print(...) to write to this console.



## Step 2: Determine the sample allocation

3

### How many plots are needed?

- Striking a balance between:
  - Staff availability
  - Budget for staff time in field
  - Desired level of accuracy
- Number and location of plots can be determined using one of the four tools on our resource page



#### Sample allocation calculator

Helps determine the number of plots needed for your study area

[DOWNLOAD CALCULATOR](#)

#### Random sampling tool

Google Earth Engine tool to determine location of plots when using a random sampling strategy

[VISIT WEBSITE](#)

#### Stratified random sampling tool

Google Earth Engine tool to help stratify a study area by ecosystem landcover class and determine location of plots using a stratified random sampling strategy

[VISIT WEBSITE](#)

#### Systemic sampling tool

Google Earth Engine tool to determine location of plots when using a systematic sampling strategy

[VISIT WEBSITE](#)

This connects to the next step...

## Step 3: Determine the sample distribution

4

### Where will plots be located?

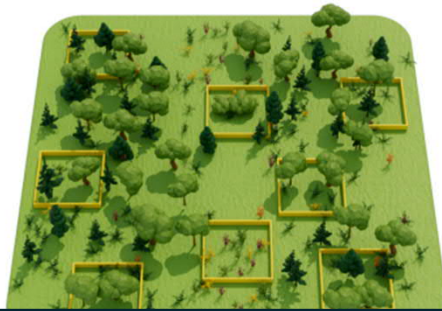
- Think back to the 4 sampling strategies
- Which works best for you?

#### CONVENIENCE/PRACTICAL SAMPLING



Want something quick and easy for a pilot season?

#### RANDOM SAMPLING



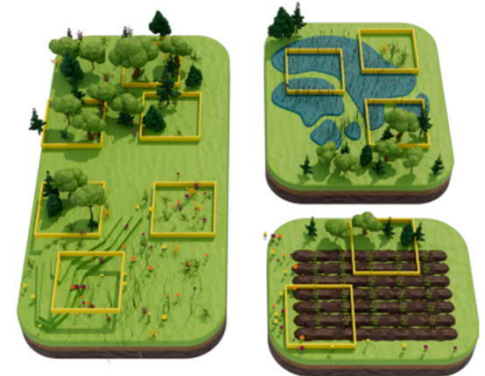
Want to remove bias while working in an area that is either uniform **or** lacking data?

#### SYSTEMATIC SAMPLING



Working in a smaller area that is largely uniform?

#### STRATIFIED-RANDOM SAMPLING



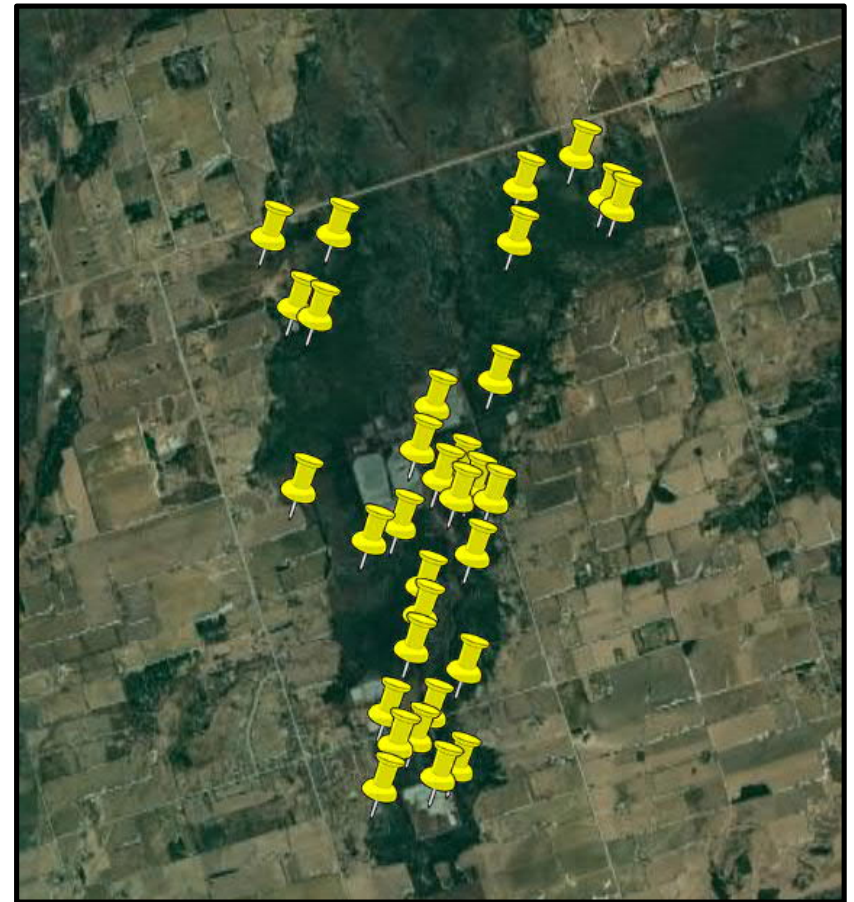
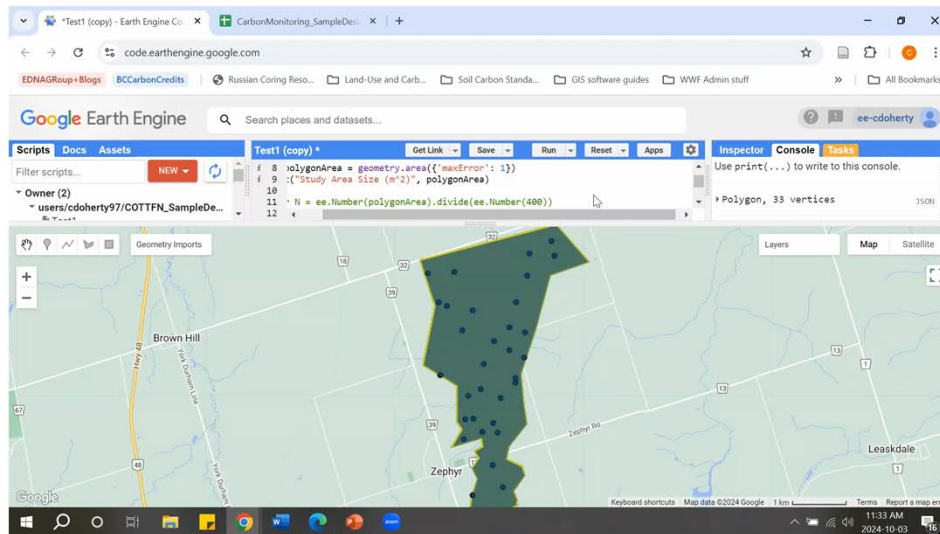
Want to accurately account for known variability across a study area?

## Step 3: Determine the sample distribution

4

GEE tools give option to export plot and sample locations

- KML file: directly upload sample points to Google Earth
- CSV file: spreadsheet with latitude and longitude coordinates



Scripts Docs Assets

- Filter scripts... NEW
- Owner (2)
    - users/cdoherty97/COTTFN\_SampleDe...
      - Test1
      - Test1 (copy)
    - users/cdoherty97/COTTFN\_Samplingd...
  - Writer
    - No accessible repositories. Click Refresh to check again.
  - Reader (1)
    - projects/AREA2/public
      - 0. Create Stratification

Test1 (copy) \*

Get Link Save Run Reset Apps

```
i 8 polygonArea = geometry.area({'maxError': 1})
i 9 :("Study Area Size (m^2)", polygonArea)
10
11 N = ee.Number(polygonArea).divide(ee.Number(400))
12 .nt("N", N)
13 r E = ee.Number()
14
15 dom sample the geometry
16
i 17 randompoints = ee.FeatureCollection.randomPoints(geometry.buffer(1), 32, 0, 1)
18 .nt(randompoints)
19 addLayer(geometry, {color: 'yellow'}, 'Study Area');
i 20 addLayer(randompoints, {color: 'black'}, 'Random Sampling Points')
21 .centerObject(geometry)
22
```

Inspector Console Tasks

Console

SUBMITTED TASKS

myExportTableTask ✓ <1m

ID: LXWJ2WATNN5J4TRHQR2HQERK

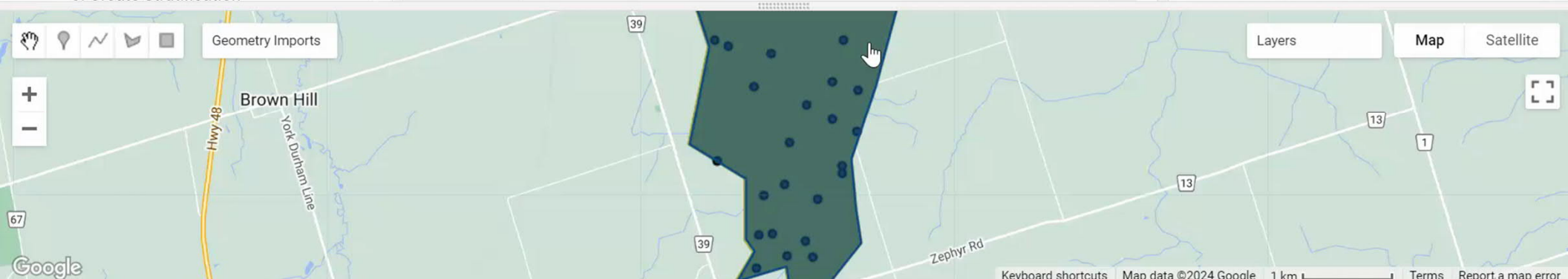
Phase: **Completed**

Runtime: **2s** (started 2024-10-03 11:34:50 -0400)

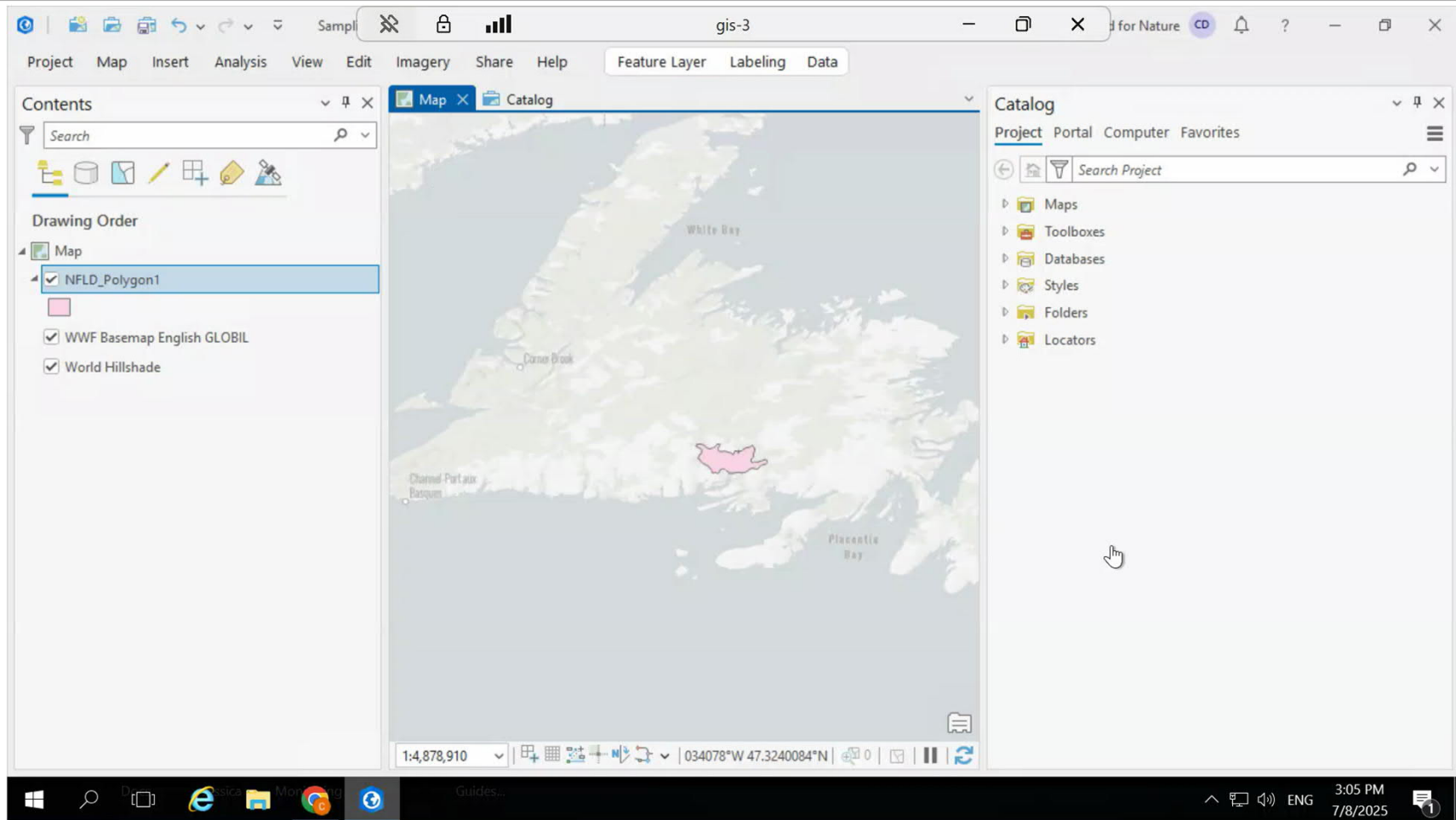
Attempted 1 time

Priority: **100** (default)

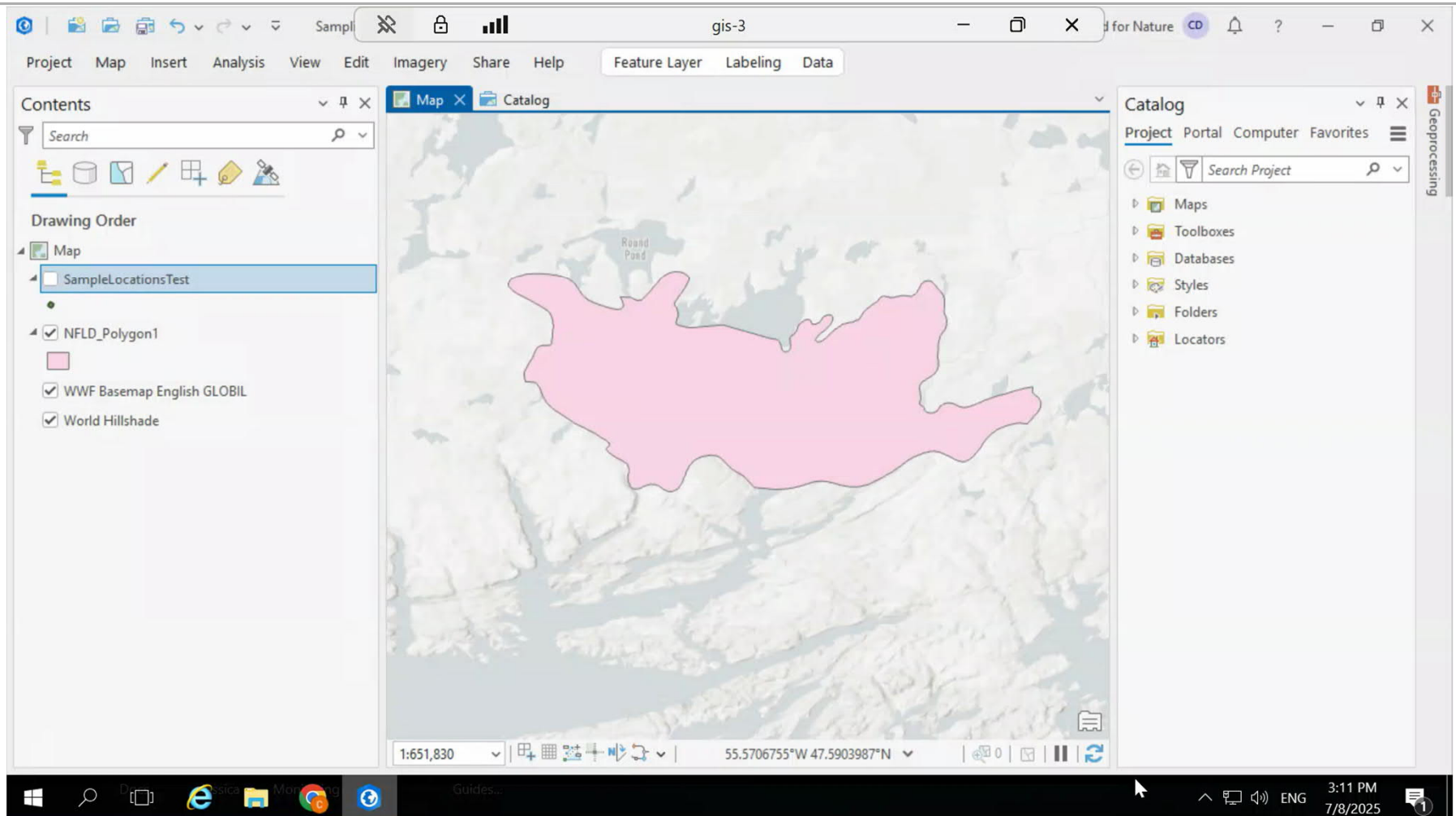
Batch compute usage: **0.0001 EECU-**



# Using ArcGIS: Random sampling



# Using ArcGIS: Systematic sampling



## Step 4: Bonus method – composite sampling

2

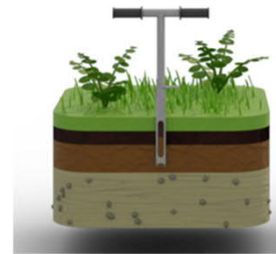
**Composite Sampling = Samples** from different locations are **pooled** together

Why?

- Increase spatial coverage of results without extra costs
- Benefits in areas with lots of variation



20 m



20 m

## Step 4: Bonus method – composite sampling

2

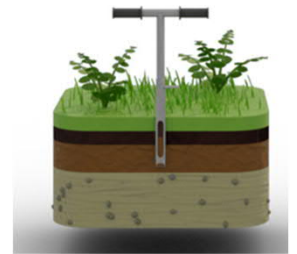
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20 m

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20 m

20 m



## Step 4: Bonus method – composite sampling

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20 m

20 m



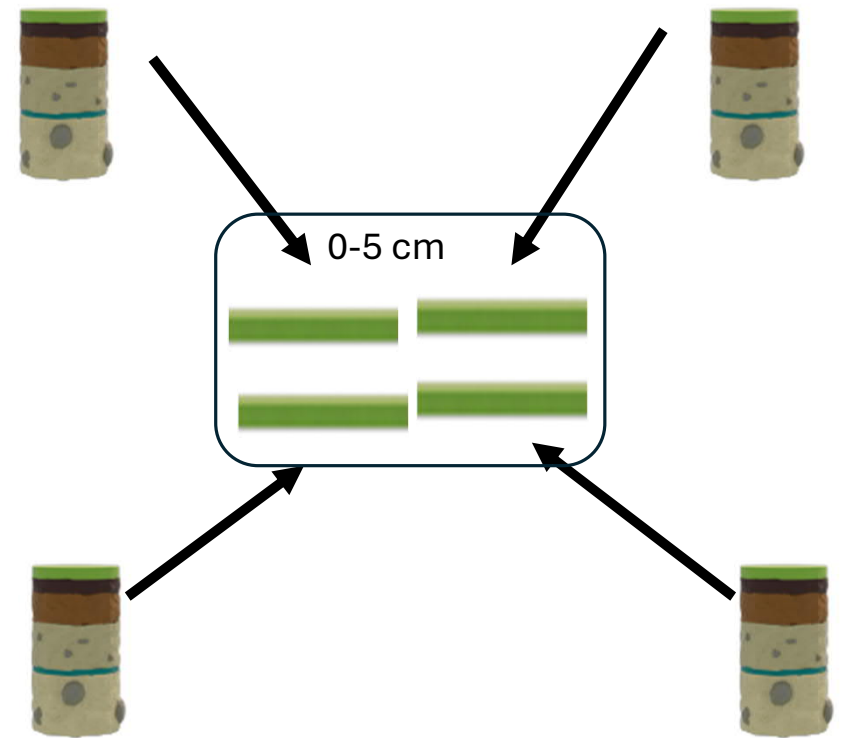
## Step 4: Bonus method – composite sampling

2

**Composite Sampling** = **Samples** from different locations are **pooled** together

How?

- Samples from **different locations**, but within the **same depth interval** are mixed together in the same sample bag



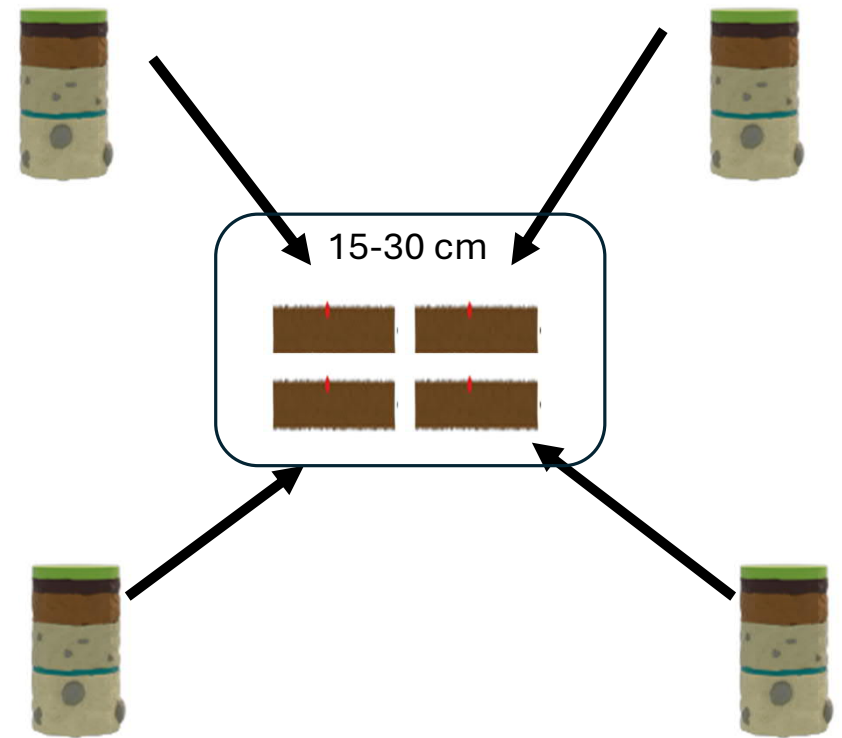
## Step 4: Bonus method – composite sampling

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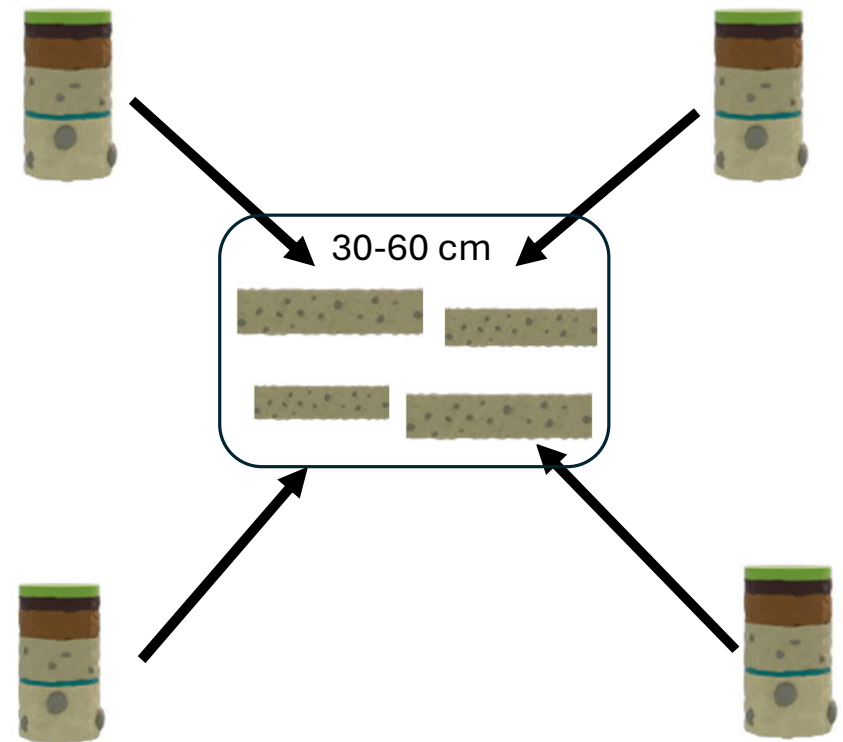
## Step 4: Bonus method – composite sampling

2

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## Step 4: Bonus method – composite sampling

2

**Composite Sampling** = **Samples** from different locations are **pooled** together

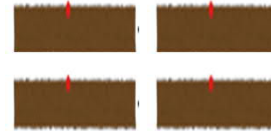
0-5 cm



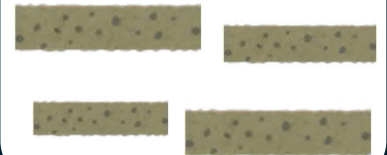
5-15 cm



15-30 cm



30-60 cm



## Step 5: Select a plot design

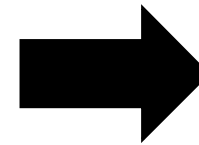
### Question 1:

What are you interested in sampling?

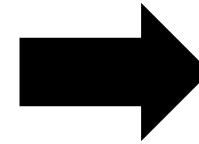
- Soils, trees, non-tree vegetation, or a combination?

### Question 2:

Will you return to the same



standalone  
vs  
integrated plots



permanent  
vs  
single-use plots

CW1

**Speaker: Clare**

Clare Wark, 2026-05-01T13:57:50.353

## Step 6: Implement field measurement protocols

6

We will cover:

- Equipment and data sheets
- Methods for sampling soils



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# Equipment

## Unique tools:

- Corer
- Core extraction tool
- Probe
- Sledgehammer
- Packaging materials
- Mini-whiteboard and marker
- Bulk density ring

## Standard tools:

- GPS
- Gloves
- Measuring tape
- Small knife
- Tarp
- Resealable bags
- Permanent markers
- Duct tape
- Measuring tape
- Shovel



CW1      **Just listing the unique equipment vs every piece of equipment here**  
Clare Wark, 2026-05-05T16:16:58.868

# Data sheets

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# Methods: Soils

CW1

3 - 4 field crew members recommended



CW1

**Speaker: [@Cathal Doherty]**

Clare Wark, 2026-05-01T13:58:23.317

# SOILS: What is soil carbon?

Soil carbon = the component of soil containing the element carbon

- **Biological** – fallen leaves, decaying plants, mosses, remnants of disturbed forest
  - Often referred to as “organic carbon”
  - Organic material typically contains 50% carbon by mass
- **Erosion/weathering** – wind and water transport tiny, mineral containing particles into an area
  - Often referred to as “inorganic carbon”



Photo: Cathal Doherty

# SOILS: What's our goal?

The aim is to collect an undisturbed vertical section of the soil profile

- **Digging soil pit** CW1
- **Open barrel corer**
  - Punch, split tube, sampling ring, PVC-corer
  - Common in dryer soils above the water table
- **Closed barrel corer**
  - Macauley/Russian peat corer or Wardenar/box-corer
  - Common in water saturated soil conditions

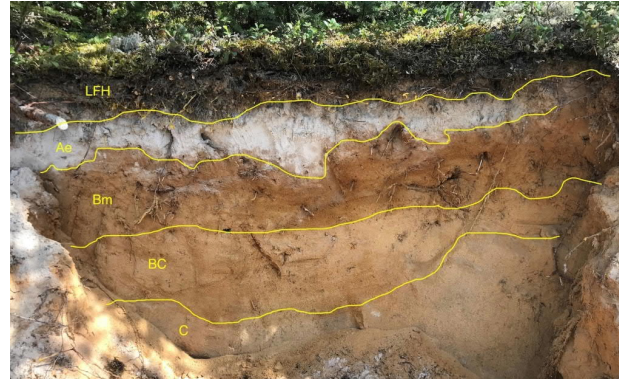


Photo: Ken Van Rees via Canadian Society of Soil Science



Photo: Cathal Doherty



Photo: Bansal et al. (2024)

CW1      **[@Cathal Doherty]** do you have a good soil pit photo we could add? If not I'll find one online  
Clare Wark, 2025-02-19T21:01:36.848

CD1 0    **The photos from NB aren't great for soil pits, I'll let you know if I find anything else**  
Cathal Doherty, 2025-02-20T12:34:32.907

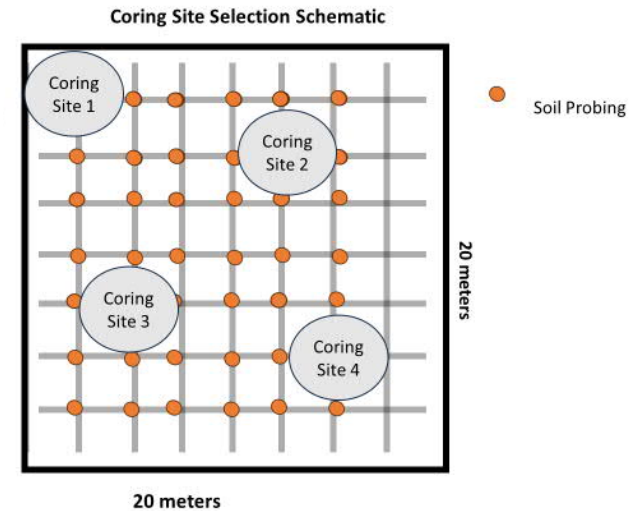
# SOILS: Carbon measurement

## 1 Site selection

- Map a 1m x 1m grid on your selected site CW1 either digitally or when you arrive onsite using a compass and measuring tape
  - At every intersection you will use a probe to measure soil depth
- Coring sites should be selected in a way that is representative of the various depths



GPS coordinates:  
Longitude: -75.69500 ° W  
Latitude: 45.4215° N  
Altitude: 260 meters  
above sea level



CW1

Cal, can you insert a slide showing (with screenshots maybe) what this actually involves?

Clare Wark, 2024-09-30T19:00:01.541

# SOILS: Carbon measurement

## 2 Preparing your site

- Document each coring site using a standardized photo series
  - 3 photos in each direction
  - 1 pointing straight down (vegetation)
  - 1 pointing straight up (canopy, if any)
- 14 photos total per coring site



# SOILS: Carbon measurement

## 2 Preparing your site

- Record the following:
  - Core ID
  - Latitude and longitude
  - Elevation
  - Any distinctive site information (e.g., previous wildfire, human disturbance, abnormal topography for study area)



# SOILS: Carbon measurement

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## Sampling techniques: Coring

CW3

- Lay down a tarp and prepare all your equipment
- Insert the corer to the desired depth, this will vary from site to site and will likely be as deep as you can go prior to hitting bedrock or hard clay
- Remove the corer and <sup>CD1</sup>place a plastic sleeve around the top end
- The core extraction tool can then be <sup>CD2</sup>used to push the sample out the top, which will be caught by the plastic sleeve

CD1      **Not PVC, maybe just plastic sleeve to generalize all soil methods**  
Cathal Doherty, 2024-03-13T15:56:18.390

CD2      **same here**  
Cathal Doherty, 2024-03-13T15:56:36.094

CW3      **Removed info on peat soils:**

For peat soils

Remove the corer and, while holding horizontally, rotate metal guard to reveal the core captured inside

A plastic tube cut lengthwise can then be placed on top

Carefully flip the tube and core upside down to release the core into the tube

Clare Wark, 2026-05-22T18:51:45.106

# SOILS: Carbon measurement

- ## 3 Sampling techniques: Coring
- Once fully removed:
    - Photograph and measure the length of the core on the tarp CW1
    - Take notes on the soil colour, transition between soil horizons, presence of larger debris or rocks, etc.
    - Record the depth of the hole you removed the core from



Photo: Clare Wark

CW1      Flag during presentation that both these measurements are important for lab analysis because it's needed to correct for accurate bulk density

Clare Wark, 2024-09-30T19:29:50.383

# SOILS: Carbon measurement

## 3 Sampling techniques: Coring

- Segmenting in the field:
    - Measure and cut the core into 5cm increments and place into labelled resealable bag
      - Smaller or larger sections may be required depending on the lab you're working with and the tests you've requested for the samples
    - Transfer to cooler and securely pack for transport
  - Segmenting later:
    - Transfer entire core into cooler and securely pack for transport
- Ensure all samples are stored in a freezer until they can be delivered to the lab

## SOILS: Carbon measurement

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### **Sampling techniques: Soil pits**

- At your site, begin digging a hole that is a minimum of 0.5m wide
- The depth of the hole will vary from site to site
  - Ensure all soil horizons are exposed
  - A photo can also be taken for your records
- Identify the soil horizons using the *Canadian System of Soil Classification*
  - This is done by looking at the colour, texture, and depth of the soil
- On your data sheet, record each soil horizon, the depth of the horizon, and details on the appearance of the soil

## SOILS: Carbon measurement

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### **Sampling techniques: Soil pits**

- Use the soil sampling ring to collect a sample in the middle of each horizon
  - This will allow the lab to determine bulk density
- Place the sample in a labelled, resealable bag and transfer to a cooler

This method is best if you intend to measure other properties beyond carbon

# SOILS: Carbon measurement

## 4 Considerations and next steps

- Ensure you have sufficient freezer space
- You will need to partner with a private lab or university/college lab to determine carbon stock, because of the equipment required
  - Discuss any requirements the lab has prior to planning in-field work
  - You can also discuss whether you'd like additional analyses completed on your samples, as this may affect sampling or storage requirements
- The results for individual cores can be scaled up to the study area
- Think ahead to where you will be storing the data provided by the lab, and how to share this with partners



CW1      Similar to comment for trees, describe how this can be done in soil context  
Clare Wark, 2024-09-30T19:43:37.182

Questions?



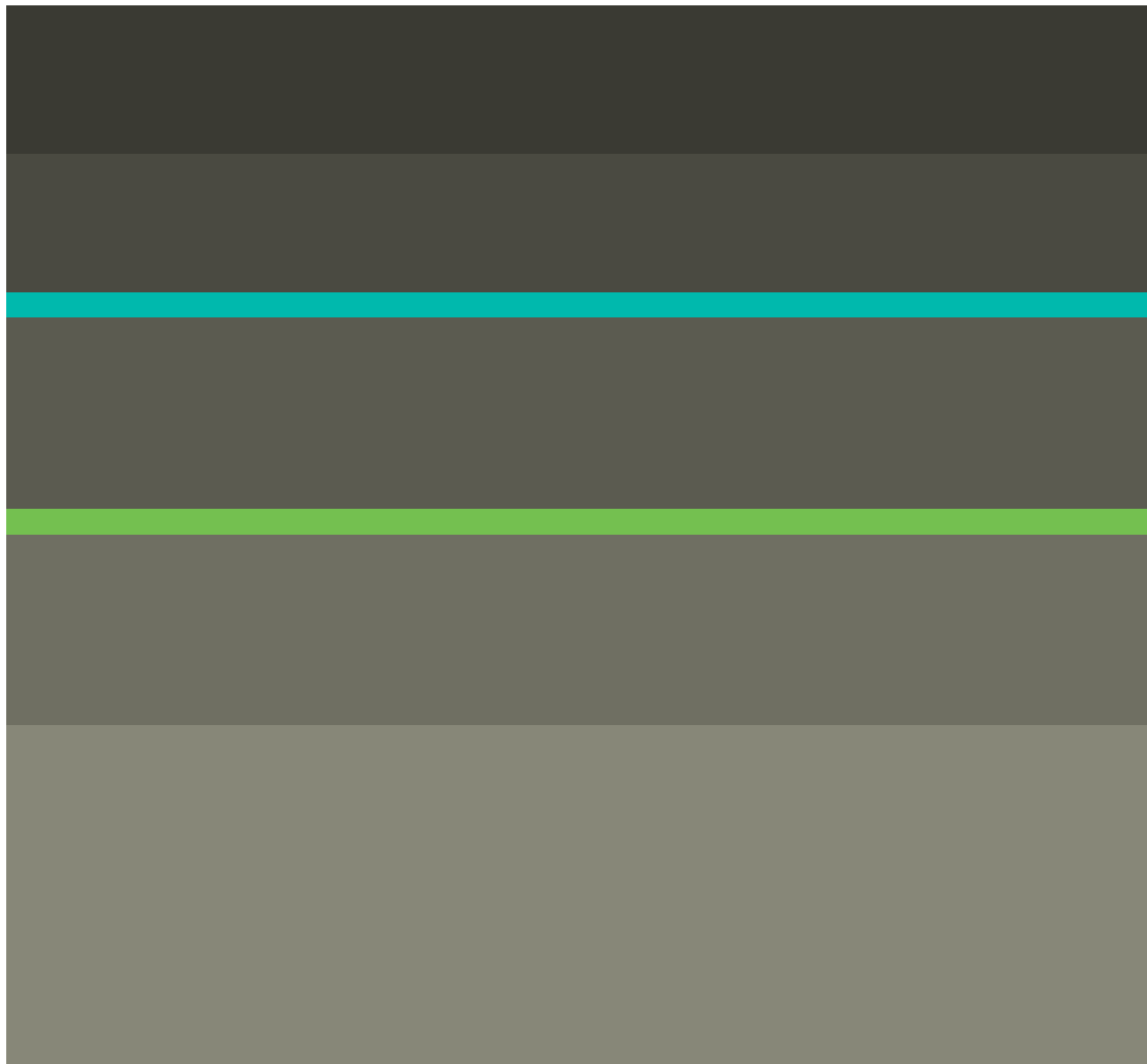
PART 2 - SECTION 06

# Data Interpretation

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From scattered cores to a continuous carbon map.

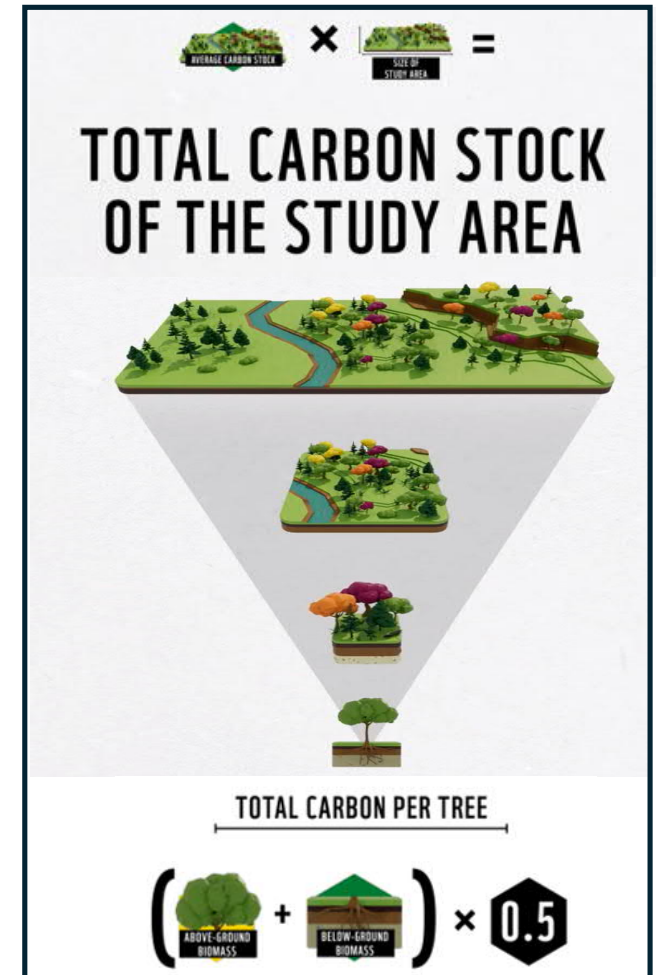
Eelgrass Coring Methods — Cathal Doherty · 24



## Step 7: Results & assessment of data

7

- Locate a lab that will analyse the samples for:
  - Wet weight
  - Dry weight
  - Organic carbon content
  - Total carbon content CW1
- Scale measurements to estimate carbon stock of the entire study area



CW1      **[@Cathal Doherty]** added ref to medium and small veg data on this slide as we discuss it in methods  
Clare Wark, 2026-05-05T17:22:13.001

## Step 7 continued: Results & assessment of data

6

Scaling up carbon stock estimates:

For each plot:

Add up carbon stock (kg) of all trees in a plot/

The size of the plot ( $\text{m}^2$ )

= Average carbon stock of the plot

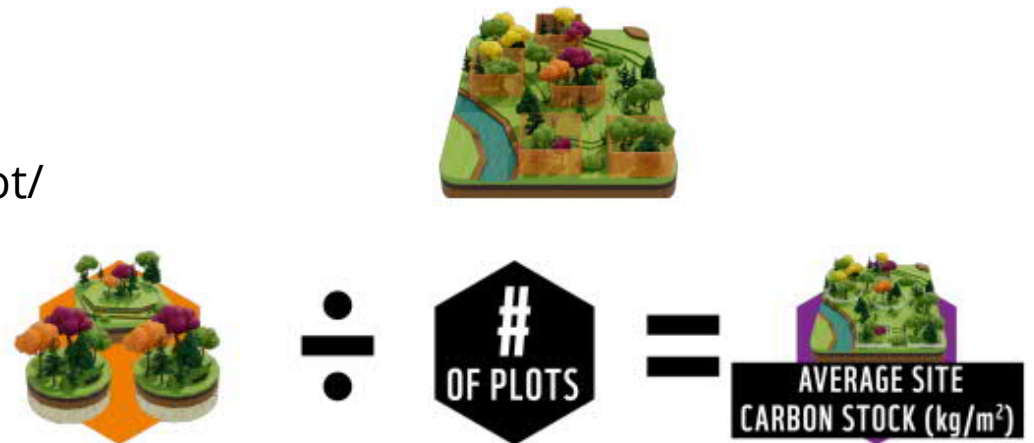


For each site:

Add up the average carbon stock for each plot/

Number of plots in the site

= Average carbon stock of the site ( $\text{kg}/\text{m}^2$ )



## Step 7 continued: Results & assessment of data

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Reporting values:

**A) Average carbon stock** of study area (**kg/m<sup>2</sup>** or **t/ha**)

Soil = Average carbon stock per area at a defined depth  
(30cm, 60cm, 100cm)

**B) Total carbon stock** (**kg** or tonnes (**t**))

Soil = Sum(average carbon stock for each site) \* Sum(size of each site)

## CALCULATING CARBON

# CORE → STUDY AREA

### CORE

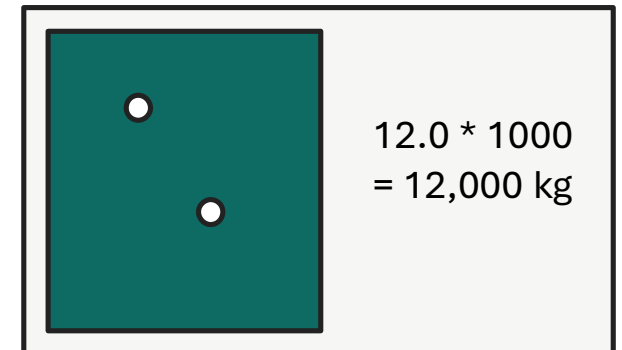
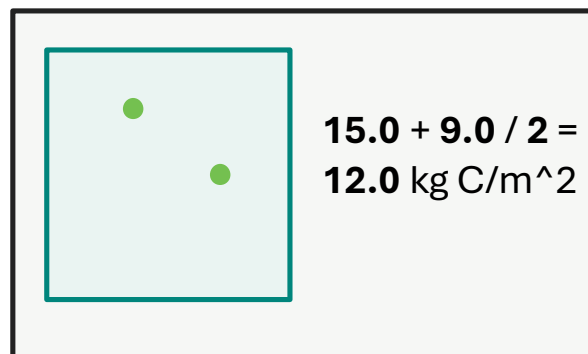
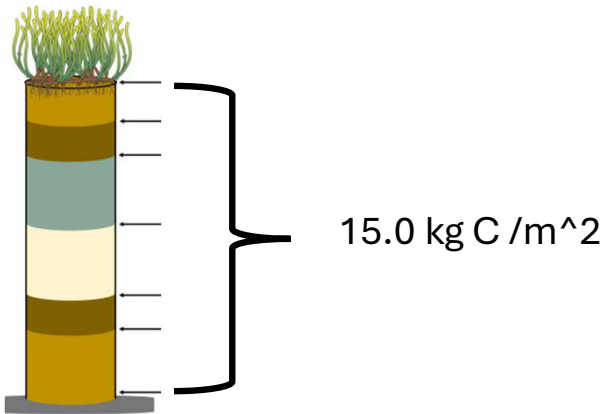
Stock of each core, in kg/m<sup>2</sup>

### SITE

Average the cores → × site area  
for the site total

### STUDY AREA

Combine sites → × total area for  
the whole stock



## BEYOND THE CORE

# Option 1 – Simple area-based extrapolation

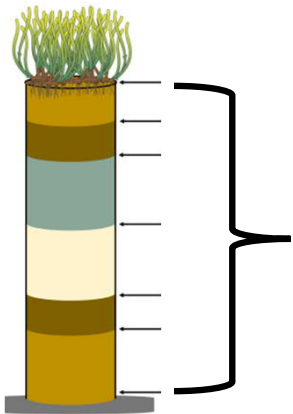
### INPUTS

Core stocks + Area

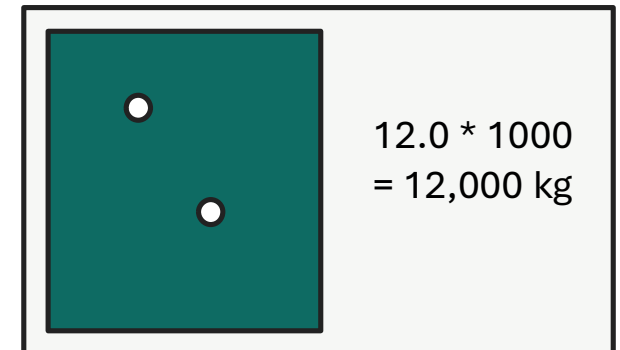
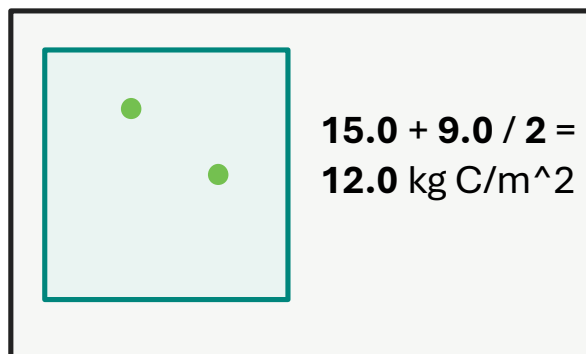


### CARBON MAP

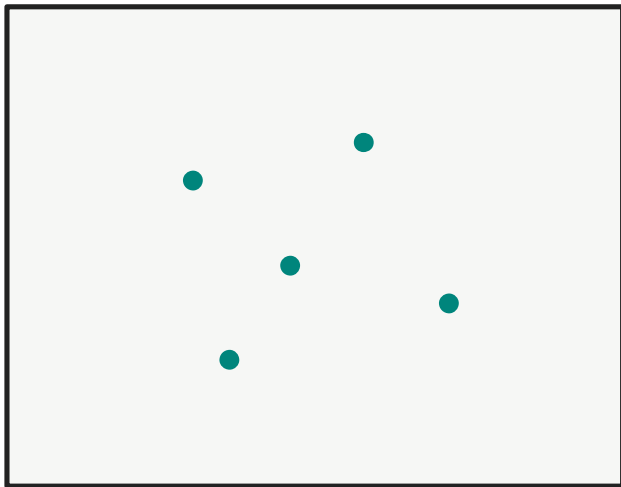
Predicted values



15.0 kg C / m<sup>2</sup>



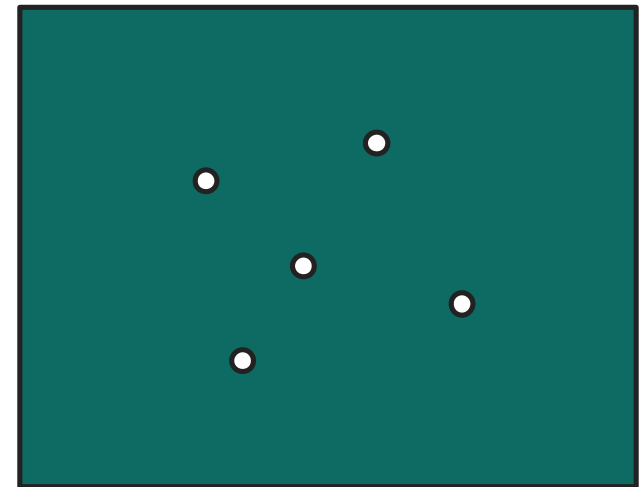
# Option 1 – Simple area-based extrapolation



**A FEW CORES**

Carbon measured at points.

Average all cores  
in an area



**A CONTINUOUS MAP**

1 value for each area averaging  
each core in the area.

## BEYOND THE CORE

# Option 2 – Modelling within areas of interest

### INPUTS

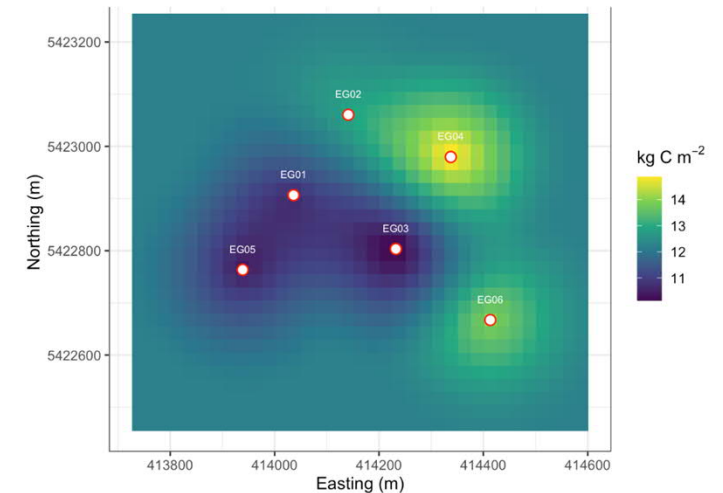
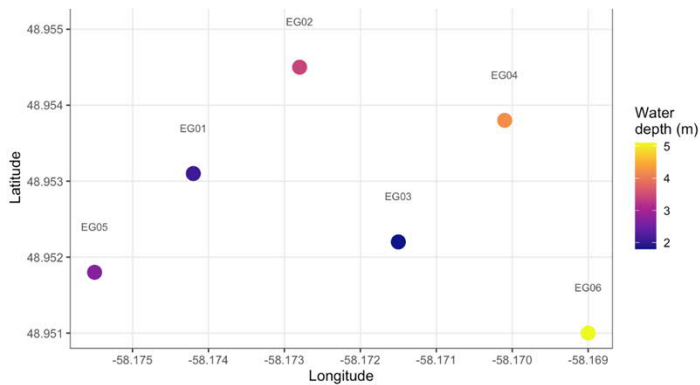
Core stocks + Area

### Extrapolation

Estimate Carbon values at every point

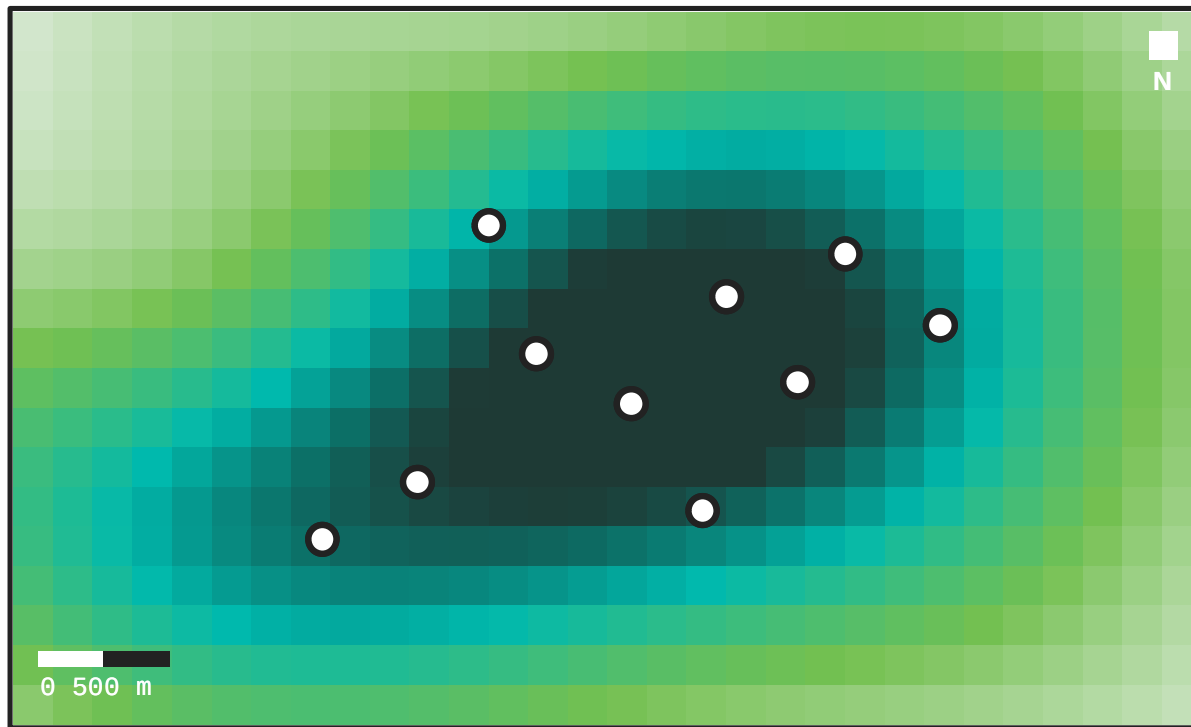
### CARBON MAP

Predicted values



BEYOND THE CORE · EXAMPLE OUTPUT

# THEMATIC CARBON MAP



Low  high Predicted carbon stock (kg C/m²)

○ Measured core locations

The model paints a predicted value into every grid cell, anchored by the real cores.

## EXPORT

Save as a **GeoTIFF** to open in **Google Earth Engine** or **ArcGIS** — for mapping, zonal totals and reporting.

Pair the map with an uncertainty layer so each estimate carries its confidence.

# Thank you!

Questions?

Email us at [science@wwfcanada.org](mailto:science@wwfcanada.org)



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